

HFT Ops Interview Preparation

Probability / Brainteasers / Code

Gabriel Kinshuk

July 27, 2026

Contents

1	Math / Probability Theory	3
1.1	Combinatorics	3
1.1.1	Permutations	3
1.1.2	Combinations	3
1.1.3	Distributing Indistinguishable Objects (Stars and Bars)	3
1.1.4	Inclusion-Exclusion	4
1.1.5	Select Worked Problems	5
1.2	Probability	6
1.2.1	Complement	6
1.2.2	Independence	7
1.2.3	Mutual Exclusivity	7
1.2.4	Select Worked Problems	7
1.3	Expected Value	10
1.3.1	Select Worked Problems	11
1.4	Conditional Probability & Bayes	15
1.4.1	Select Worked Problems	17
1.5	Distributions	19
1.5.1	Bernoulli Distribution	20
1.5.2	Binomial Distribution	20
1.5.3	Geometric Distribution	21
1.5.4	Negative Binomial Distribution	22
1.5.5	Poisson Distribution	22
1.5.6	Exponential Distribution	23
1.5.7	Uniform Distribution	24
1.5.8	Normal (Gaussian) Distribution	24
1.5.9	Lognormal Distribution	26
1.5.10	Reference Tables	26
1.5.11	Select Worked Problems	28
1.6	Markov Chains	30
1.6.1	Absorbing Chains and First-Step Analysis	30
1.6.2	Recurrent Chains and the Stationary Distribution	31
1.6.3	Select Worked Problems	32
2	Brainteasers	34
2.1	Strategies	34
2.1.1	Framing	34
2.1.2	Core Probabilistic Reductions	34

2.1.3	Combinatorial and Logical Tactics	35
2.1.4	Adversarial and Sequential Decisions	35
2.1.5	Counting Information	36
2.1.6	Estimation	36
2.1.7	Verification	37
2.2	Select Worked Problems	38
3	Code	50
3.1	SQL Patterns (PostgreSQL)	50
3.1.1	Filtering and Joins	50
3.1.2	Aggregation	51
3.1.3	NULL Handling	52
3.1.4	Window Functions	52
3.1.5	Deduplication	54
3.1.6	String Manipulation	55
3.1.7	Command Reference	57
3.2	Python Patterns	60
3.3	Algorithmic Paradigms	67
3.4	Recursion and Memoization	68
3.5	Heaps and Priority Queues	71
3.6	Sliding Window	72

1 Math / Probability Theory

1.1 Combinatorics

A counting problem is pinned down by two questions. Does the order of the chosen elements matter, and may an element be chosen more than once. The formulas below are organized around those two axes, plus a correction for identical elements and two escape hatches (the complement and inclusion-exclusion) for when a direct count fans out into many cases.

1.1.1 Permutations

Definition 1.1 (Permutation). A permutation is an ordered selection. Choosing k items from n distinct options where order matters gives

$$P(n, k) = \frac{n!}{(n - k)!}.$$

The formula counts position by position. The first slot has n candidates, the second has $n - 1$ once one item is placed, and so on for k slots, giving the falling product $n(n - 1) \cdots (n - k + 1)$. Writing that product as $\frac{n!}{(n - k)!}$ just cancels the unused tail $(n - k)!$.

Definition 1.2 (Permutations with identical elements). When the n items being arranged include groups of identical copies with frequencies $n_1, n_2, \dots, n_k$, the number of distinguishable arrangements is

$$\frac{n!}{n_1! n_2! \cdots n_k!}.$$

Start from the $n!$ arrangements that would exist if every item were distinct. Each identical group of size n_i has $n_i!$ internal orderings that all produce the same visible arrangement, so dividing by each $n_i!$ removes exactly those duplicates.

1.1.2 Combinations

Definition 1.3 (Combination). A combination is an unordered selection. Choosing k items from n distinct options where order does not matter gives

$$C(n, k) = \binom{n}{k} = \frac{n!}{k! (n - k)!}.$$

Combinations come from permutations by discarding the ordering. Each unordered set of k items can be arranged in $k!$ ways, and $P(n, k)$ counts all of those as distinct, so

$$C(n, k) = \frac{P(n, k)}{k!}.$$

1.1.3 Distributing Indistinguishable Objects (Stars and Bars)

Theorem 1.4 (Stars and bars). *The number of ways to place n identical objects into k distinct bins, where a bin may hold any number of objects including zero, is*

$$\binom{n + k - 1}{k - 1} = C(n + k - 1, k - 1).$$

Proof. Lay the n objects in a row as stars and insert $k - 1$ dividers (bars) to cut the row into k groups. Reading left to right, objects before the first bar fall in bin 1, objects between the first and second bar fall in bin 2, and so on through bin k . Every placement of the bars produces one distribution, and every distribution corresponds to exactly one placement. The row has $n + (k - 1)$ positions, and a distribution is fixed by choosing which $k - 1$ of them hold bars, which is $\binom{n+k-1}{k-1}$. $\square$

1.1.4 Inclusion-Exclusion

Theorem 1.5 (Inclusion-Exclusion). *The size of a union is the alternating sum of the sizes of all intersections. For two and three sets,*

$$|A \cup B| = |A| + |B| - |A \cap B|,$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

The mechanism is bookkeeping on how many times each element gets counted. Summing the individual set sizes counts an element that lies in two sets twice, so subtracting the pairwise intersections corrects it back to one. An element in all three sets is now counted $3 - 3 = 0$ times, so adding the triple intersection restores it to one. The pattern continues to any number of sets: terms over an odd number of sets are added and terms over an even number are subtracted, which leaves every element counted exactly once no matter how many sets contain it.

Which counting tool

Order matters?	Repeats allowed?	Count
yes	no	$P(n, k) = \frac{n!}{(n-k)!}$
no	no	$C(n, k) = \binom{n}{k}$
yes	yes	n^k
no	yes	$\binom{n+k-1}{k-1}$ (stars and bars)

Two escape hatches handle counts that fan out into many subcases. Use the **complement** ($1 - P$ of the unwanted case) on “at least one” phrasing, and use **inclusion-exclusion** on “divisible by a or b ” style overlaps.

1.1.5 Select Worked Problems

Problem 1.1

How many ways can you arrange the letters in the word BANANA?

Solution. There are $n = 6$ elements. Repeated items have frequency 3 and 2 (Ns and As). Then the number of arrangements are $\frac{6!}{3! \times 2!} = \frac{720}{12} = 60$.

Problem 1.2

You roll 2 dice. How many distinct outcomes are there? What if you can't tell the dice apart?

Solution. With two dice, there are $6 \times 6 = 36$ outcomes if we can tell the dice apart. If we *can't* tell the dice apart, then we can solve either logically or formulaically:

Logically, we can see that from the original 36 combinations, there are 6 outcomes where the dice are the same, leaving 30 outcomes using mixed pairs of dice. Of these, half are duplicate when we can't tell the dice apart, leaving 15 outcomes. We can then add the 6 double outcomes back in for a total of 21 outcomes.

Formulaically, we can solve using a similar approach, taking the sum of mixed pair outcomes and double outcomes as $C(6, 2) + C(6, 1) = 15 + 6 = 21$.

Lastly, we can consider this problem using the stars and bars approach, where we notice that there are $n = 2$ identical dice and 6 distinct bins with $k = 5$ dividers between them. Then we can count the outcomes by $C(2 + 6 - 1, 6 - 1) = C(7, 5) = 21$

Problem 1.3

How many ways can you distribute 8 identical balls into 3 distinct boxes?

Solution. This is a direct application of stars and bars with $n = 8$ and $k = 3$. The number of outcomes is $C(8 + 3 - 1, 3 - 1) = C(10, 2) = \frac{10 \times 9 \times 8 \times \dots \times 3}{2} = 45$.

Problem 1.4

A deck has 52 cards. How many 5-card hands contain at least one ace?

Solution. Since this solution asks for the number of hands containing at least one ace, it will be simpler to compute the complement. This is the number of hands containing no aces, we then subtract from the total number of 5 card hands: $C(52, 5) - C(48, 5) = 886,656$.

Problem 1.5

You have 5 books on a shelf. In how many ways can you choose 3 books and arrange them left-to-right on a new shelf?

Solution. This is simply $C(5, 3) \times P(3, 3) = \frac{5 \times 4 \times 3}{3!} \times \frac{3!}{0!} = \frac{60 \times 3!}{3!} = 60$.

Problem 1.6

A pizza place offers 8 toppings. How many different pizzas can you make if you can choose any number of toppings (including none)?

Solution. We can solve this by taking the summation of $C(8, i)$ from $i = 0$ to $i = 8$:

$$Total\ outcomes = \sum_{i=0}^8 C(8, i) = 256$$

There is a more succinct solution we can find if we approach the problem seeing each topping as the binary choice of being either on or off of the pizza, then the total outcomes are $2^8 = 256$.

Problem 1.7

How many ways can 12 people be split into 3 groups of 4?

Solution. We choose each group independently with the remaining members of the pool, and then divide by the reorderings since the groups are indistinguishable:

$$\frac{C(12, 4) \times C(8, 4) \times C(4, 4)}{3!} = \frac{495 \times 70 \times 1}{6} = 34,650.$$

1.2 Probability

Definition 1.6 (Sample Space). The sample space Ω is the set of all possible outcomes of an experiment.

Most problems begin by fixing the sample space and then identifying the subset of outcomes that count as favourable. Getting the sample space right, which means choosing the correct unit of observation and the correct notion of “equally likely”, settles most of the difficulty before any arithmetic starts.

Definition 1.7 (Event). An event is a subset of the sample space, the collection of outcomes of interest. Its probability is the total weight of the outcomes it contains.

1.2.1 Complement

Complement

For any event A ,

$$P(A) = 1 - P(A^c),$$

where A^c is the event that A does not occur.

The complement is the first move whenever the direct event splits into many cases while its negation is a single clean case. “At least one” is the signature cue: the direct count runs over one, two, three or more successes, while the negation is the single event of zero successes. The birthday and “at least one head” problems below both collapse this way.

1.2.2 Independence

Definition 1.8 (Independence). Two events are independent when the occurrence of one leaves the probability of the other unchanged. Formally,

$$P(A \cap B) = P(A) P(B).$$

The multiplication rule is the practical test. Drawing *with* replacement keeps every draw independent because the pool is restored each time. Drawing without replacement makes the draws dependent, since removing an item changes both the size and the composition of the pool for the next draw. Independence is a statement about probabilities and is a separate idea from mutual exclusivity, which is a statement about outcomes.

1.2.3 Mutual Exclusivity

Definition 1.9 (Mutual exclusivity). Two events are mutually exclusive when they cannot both occur, so $A \cap B = \emptyset$. For such events the union probability adds:

$$P(A \cup B) = P(A) + P(B).$$

This is the special case of inclusion-exclusion where the intersection is empty. The general identity $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ carries the overlap term $P(A \cap B)$, and mutual exclusivity sets that term to zero. A common error is to add probabilities for events that can co-occur, which double counts the overlap.

1.2.4 Select Worked Problems

Problem 1.8

How many people do you need in a room for a >50% chance that two share a birthday?

Solution. Computing this directly is impractical, since there are many ways for multiple people to share a birthday. Instead, using the complement, we simply subtract the probability that *no* people share a birthday from 1:

$$P(\text{all different}) = \prod_{i=1}^n \frac{365 - i + 1}{365}$$

We multiply additional factors until the quantity drops *below* 0.5 (because then $1 - P$ is greater than 0.5). This turns out to be $n = 23$, where $P(\text{all different}) \approx 0.493$ and thus $P(\text{birthday is shared}) \approx 0.507$.

Problem 1.9

You flip a fair coin 3 times. What's the probability of getting exactly 2 heads?

Solution. Flipping a coin three times has eight outcomes. Of these, three involve exactly two heads, thus the probability is $\frac{3}{8} = 0.375$.

If we don't want to enumerate each of the eight options and count directly, we could use $C(3, 2) = 3$ to find the number of outcomes involving two heads.

Problem 1.10

Roll two dice. What's the probability the sum is 7?

Solution. There are 36 outcomes of rolling two dice. Of these, the pairs that equal 7 are $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$. Then the probability is $\frac{6}{36} \approx 0.167$.

Problem 1.11

You draw 2 cards from a standard 52-card deck (without replacement). What's the probability both are hearts?

Solution. Without replacement, the draws are dependent. Then, the probability is:

$$\frac{13}{52} \times \frac{12}{51} \approx 0.0588$$

Note that *both* the numerator and the denominator shrank after we removed the first card.

We could also solve this by counting, if we divide the number of ways to pick two hearts by the total number of ways to pick two cards:

$$\frac{C(13, 2)}{C(52, 2)} = \frac{78}{1326} \approx 0.0588$$

Problem 1.12

You flip a fair coin 5 times. What's the probability of getting at least one head?

Solution. We can use the complement, the probability of getting *no* heads, subtracted from 1:

$$1 - \frac{1}{2^5} = 0.96875$$

Problem 1.13

You roll a fair die twice. What's the probability the second roll is strictly greater than the first?

Solution. There are 36 total ways to roll two dice. If the first roll is a 1, then there are 5 ways to roll higher on the second die. Similarly, there are 4 ways to roll higher if we rolled a 2, and so on.

Thus the number of favourable outcomes $1 + 2 + \cdots + 5$ or the triangle number of 5, $T_5 = 15$. The final answer is:

$$\frac{15}{36} \approx 0.417$$

Alternatively, we could solve by symmetry, calling roll 1 and 2 as A and B , respectively. If we notice that $P(A > B) = P(B < A)$, and $P(A = B) = \frac{6}{36}$, then we evenly divide the remaining outcomes by 2 to get $\frac{15}{36}$.

Problem 1.14

A box has 5 white and 3 black balls. You draw 3 without replacement. What's the probability of drawing exactly 2 white and 1 black?

Solution. The total ways to draw 3 balls are $C(8, 3) = 56$. We then find the ways to choose 2 white balls and 1 black ball individually, which is $C(5, 2)$ and $C(3, 1)$, then the answer is:

$$\frac{C(5, 2) \times C(3, 1)}{C(8, 3)} = \frac{10 \times 3}{56} \approx 0.536$$

Since we remove the balls without replacement, this problem follows the hypergeometric distribution.

Problem 1.15

You randomly pick a number from 1 to 100. What's the probability it's divisible by 3 or 5?

Solution. There are 33 numbers divisible by 3, and 20 numbers divisible by 5. The numbers divisible by both are the multiples of 15 of which there are 6 (15, 30, 45, 60, 75, 90). Thus:

$$P_{Div}(A \cup B) = \frac{33}{100} + \frac{20}{100} - \frac{6}{100} = 0.47$$

Problem 1.16

A family has 3 children. Assuming each child is equally likely to be a boy or girl, what's the probability that all 3 are the same gender?

Solution. There are eight outcomes, of those there are two in which all children are the same gender, so the answer is $\frac{2}{8} = 0.25$.

Problem 1.17

You throw 3 dice. What's the probability ALL three show different numbers?

Solution. There are $6^3 = 216$ total ways to throw three dice. After we throw the first dice, there are 5 options remaining, and 4 after the second dice. Then the answer is:

$$\frac{6 \times 5 \times 4}{6^3} = \frac{120}{216} \approx 0.556$$

Problem 1.18

You toss a fair coin until you get heads. What's the probability you need exactly 3 tosses?

Solution. We stop at the first heads, so needing exactly 3 tosses means the first two tosses are tails and the third is heads: T, T, H. Since tosses are independent and each outcome has probability $\frac{1}{2}$,

$$P(\text{exactly 3}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}.$$

We do not enumerate anything else, because exactly one sequence produces this event.

Remark (Two ways probability gets counted). The reason we can pin down this single case without listing every outcome is that this problem lives in a different kind of sample space than counting problems like the birthday question.

When every outcome is equally likely, probability is combinatorial. You divide favorable outcomes by total outcomes, so you have to characterize the entire sample space. Each outcome carries no probability on its own and earns weight only by being counted against the total.

When outcomes are not equally likely, each outcome carries its own probability directly, usually a product of independent step probabilities. Here the sequence TTH has probability $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$ by itself, with no denominator to divide by. That is why one case can be solved in isolation.

These are the same axioms applied to different spaces. Counting favorable over total is the special case that appears when all per-outcome probabilities are equal, so the sum of probabilities collapses into a plain count. The coin problem is the general version, where the per-outcome probabilities differ and you take the one you want.

Problem 1.19

In a room of 5 people, what's the probability that at least two share a birthday? (Assume 365 days, equally likely.)

Solution. Again, we use the complement to find 1 minus the probability that no one shares a birthday:

$$1 - \left(\frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \frac{362}{365} \times \frac{361}{365} \right) \approx 1 - 0.9729 \approx 0.0271$$

1.3 Expected Value

Definition 1.10 (Expected Value). The expected value of a random variable is the probability-weighted average of its outcomes:

$$\mathbb{E}[X] = \sum_i x_i P(X = x_i).$$

It is the long-run mean over many repetitions. The expected value need not be an attainable outcome (a fair die has $\mathbb{E}[X] = 3.5$) and need not be the most likely one.

Linearity of expectation

For any random variables X and Y and any scalars a, b ,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

This holds with no independence assumption. Paired with indicator variables it is the highest-yield tool for expectation brainteasers, since it turns a hard expected count into a sum of easy probabilities.

Definition 1.11 (Indicator Random Variable). An indicator $\mathbf{1}\{A\}$ equals 1 when event A occurs and 0 otherwise. Its expectation is the event's probability,

$$\mathbb{E}[\mathbf{1}\{A\}] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A).$$

The variable itself is trivial. The leverage is the decomposition it enables. To find the expected *count* of how many events in a family occur, write that count as a sum of indicators, one per event, then apply linearity:

$$\mathbb{E}\left[\sum_i \mathbf{1}\{A_i\}\right] = \sum_i \mathbb{E}[\mathbf{1}\{A_i\}] = \sum_i P(A_i).$$

Because linearity ignores dependence, this works even when the events are entangled, which is what collapses an otherwise messy count into a sum of easy probabilities. The shuffle and fixed-point problems below are the standard instances.

Theorem 1.12 (Law of Total Expectation). *If the cases $B_1, \dots, B_n$ partition the sample space, meaning they are mutually exclusive and cover every outcome, then*

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X \mid B_i] \cdot P(B_i).$$

This regroups the definition of expected value by cases. Rather than averaging over individual outcomes, average the conditional mean inside each case and weight it by the case probability. Expanding each $\mathbb{E}[X \mid B_i]$ recovers the flat sum, so the identity adds no new information, but it changes what is computable. It is the tool of choice in three situations:

- you are handed conditional averages and case probabilities rather than the full outcome distribution
- X is continuous, so the flat sum $\sum_x x P(X = x)$ does not apply while conditioning on discrete cases still does
- $\mathbb{E}[X \mid B_i]$ refers back to $\mathbb{E}[X]$ itself, which turns the conditioning into a solvable equation

The cases must be disjoint and exhaustive. Overlapping or missing cases break the weighting.

Example 1.13 (Conditioning on cases). A commute time depends on the weather. Sunny days are 60% of days with a 20 minute average, and rainy days are 40% with a 35 minute average. The two cases partition every day, so

$$\mathbb{E}[\text{commute}] = 0.6(20) + 0.4(35) = 12 + 14 = 26 \text{ minutes.}$$

Example 1.14 (Recursive expectation). Take the expected number of rolls of a die until the first 6, and call it E . One roll always happens. With probability $\frac{1}{6}$ it is a 6 and the process stops, and with probability $\frac{5}{6}$ it is not, which leaves the same problem ahead with the same expectation E :

$$E = 1 + \frac{5}{6}E \longrightarrow \frac{1}{6}E = 1 \longrightarrow E = 6.$$

This is the mean of the geometric distribution, the expected number of trials until the first success, which is $\frac{1}{p}$ with $p = \frac{1}{6}$.

1.3.1 Select Worked Problems

Problem 1.20

A game costs \$1. You flip a coin: heads you win \$3, tails you win nothing. What's $\mathbb{E}[\text{Profit}]$?

Solution. $EV = 0.5 \cdot 2 + 0.5 \cdot -1 = +0.5$

Problem 1.21

Roll a die. If it shows 6, you roll again and add. Otherwise, your score is the number shown. What's the expected score?

Solution.

$$EV = 3.5 + \frac{1}{6} \cdot EV \longrightarrow \frac{5}{6} \cdot EV = 3.5 \longrightarrow EV = 4.2$$

Problem 1.22

Shuffle a standard deck. What's the expected number of cards that are in their "correct" position (card k is in position k)?

Solution. This is linearity of expectation applied to indicators. Let $I_k = 1$ if card k lands in position k , and 0 otherwise. In a random shuffle each card is equally likely to occupy any of the 52 positions, so $\mathbb{E}[I_k] = P(I_k = 1) = \frac{1}{52}$. The number of correctly placed cards is $M = \sum_{k=1}^{52} I_k$, and by linearity

$$\mathbb{E}[M] = \sum_{k=1}^{52} \mathbb{E}[I_k] = 52 \cdot \frac{1}{52} = 1.$$

The I_k are dependent, but linearity does not require independence, so that never matters.

Problem 1.23

You flip a coin until you get heads. How many flips do you expect?

Solution.

$$EV = 1 + 0.5 \cdot EV \longrightarrow 0.5 \cdot EV = 1 \longrightarrow EV = 2$$

Problem 1.24

A cereal box contains 1 of 4 different toys (equally likely). How many boxes must you buy to expect to collect all 4?

Solution. The total number of boxes is a sum of four phase waits, so we use linearity of expectation. In phase k we already hold $k - 1$ toys and buy boxes until a new one appears. The per-trial success probability is $p = \frac{\text{favourable}}{\text{total}}$, the number of missing toys over 4. We want the reciprocal, the expected number of trials per success, which is $\frac{1}{p}$:

$$\begin{aligned} 0 \text{ toys, } p &= \frac{4}{4} \longrightarrow \mathbb{E}[T_1] = \frac{1}{p} = 1 \\ 1 \text{ toy, } p &= \frac{3}{4} \longrightarrow \mathbb{E}[T_2] = \frac{1}{p} = \frac{4}{3} \\ 2 \text{ toys, } p &= \frac{2}{4} \longrightarrow \mathbb{E}[T_3] = \frac{1}{p} = 2 \\ 3 \text{ toys, } p &= \frac{1}{4} \longrightarrow \mathbb{E}[T_4] = \frac{1}{p} = 4 \end{aligned}$$

$$\mathbb{E}[T] = \mathbb{E}[T_1] + \mathbb{E}[T_2] + \mathbb{E}[T_3] + \mathbb{E}[T_4] = 1 + \frac{4}{3} + 2 + 4 = \frac{25}{3} \approx 8.33$$

Problem 1.25

I roll a die repeatedly and keep a running sum. What's the expected number of rolls until the sum first reaches or exceeds 10?

Solution. We solve recursively. Let $\mathbb{E}[s]$ be the expected number of additional rolls needed to first reach a sum of at least 10, starting from current sum s . The answer we want is $\mathbb{E}[0]$.

Base case: $\mathbb{E}[s] = 0$ for all $s \geq 10$, since the target is already met. For $s < 10$, one roll costs 1 and moves to $s + 1, \dots, s + 6$ with equal probability $\frac{1}{6}$:

$$\mathbb{E}[s] = 1 + \frac{1}{6} \sum_{j=1}^6 \mathbb{E}[s + j], \quad \mathbb{E}[s] = 0 \text{ for } s \geq 10.$$

Working downward from the base case:

$$\begin{aligned} \mathbb{E}[9] &= 1 + \frac{1}{6}(\mathbb{E}[10] + \dots + \mathbb{E}[15]) = 1 \\ \mathbb{E}[8] &= 1 + \frac{1}{6}\mathbb{E}[9] = \frac{7}{6} \\ \mathbb{E}[7] &= 1 + \frac{1}{6}(\mathbb{E}[8] + \mathbb{E}[9]) = \frac{49}{36} \\ &\vdots \\ \mathbb{E}[0] &\approx 3.6 \end{aligned}$$

Problem 1.26

You roll a fair six-sided die. After seeing the result, you may keep it as your score, or reroll. If you reroll, the new value replaces the old one (you cannot go back to a discarded roll). You are allowed up to three rolls total, and your score is the value you are holding when you stop. Playing optimally to maximize your expected score, what is that expected score, and what is the stopping rule at each stage?

Solution. Let $\mathbb{E}[r]$ be the expected score with r rerolls remaining. Solve backwards.

With no rerolls left, you keep whatever you roll, so $\mathbb{E}[0] = 3.5$.

With one reroll left, you keep the current face only if it beats the continuation value $\mathbb{E}[0] = 3.5$, i.e. keep on $\{4, 5, 6\}$ and reroll otherwise:

$$\mathbb{E}[1] = \frac{1}{6}(4) + \frac{1}{6}(5) + \frac{1}{6}(6) + \frac{3}{6}(3.5) = \frac{17}{4} = 4.25.$$

With two rerolls left, the continuation value is now $\mathbb{E}[1] = 4.25$, so you keep only faces that beat it, $\{5, 6\}$, and reroll on $\{1, 2, 3, 4\}$:

$$\mathbb{E}[2] = \frac{1}{6}(6) + \frac{1}{6}(5) + \frac{4}{6}(4.25) = \frac{14}{3} \approx 4.67.$$

Problem 1.27

You pick a random number uniformly from $[0, 1]$. Then you pick another. Then another. You stop when the running total exceeds 1. How many numbers do you expect to pick?

Solution. Let N be the number of draws until the running sum first exceeds 1, and let $X_1, X_2, \dots$ be the draws. We want $\mathbb{E}[N]$.

Step 1: a tail probability. The *tail probability* $\mathbb{P}(N > n)$ is the chance N exceeds a cutoff n , i.e. the chance we still need more draws after n of them. Needing more than n draws means the first n have not yet crossed 1:

$$N > n \iff X_1 + \dots + X_n \leq 1.$$

Step 2: geometry The point $(X_1, \dots, X_n)$ lands uniformly in the unit hypercube $[0, 1]^n$, which has volume 1. Since the draws are uniform, the probability of landing in any region equals that region's volume. The region where the coordinates are nonnegative and sum to at most 1 is a *simplex* (the n -dimensional triangle: a segment in 1D, triangle in 2D, tetrahedron in 3D). Its volume is $\frac{1}{n!}$ ($V_n(t) = \int_0^t V_{n-1}(t-x) dx$, starting from $V_1(t) = t$. Then $V_n(t) = \frac{t^n}{n!}$, evaluated at $t = 1$, $V_n(1) = \frac{1}{n!}$), so

$$\mathbb{P}(N > n) = \mathbb{P}(X_1 + \dots + X_n \leq 1) = \frac{1}{n!}.$$

Step 3: sum the tails. For a nonnegative integer variable, the mean equals the sum of its tail probabilities:

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \mathbb{P}(N > n).$$

This holds because an outcome $N = k$ clears the cutoffs $n = 0, 1, \dots, k-1$, so it is counted exactly k times across the sum, contributing its own value. Substituting the tails:

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \dots = e \approx 2.718.$$

1.4 Conditional Probability & Bayes

Definition 1.15 (Conditional Probability). The probability of A given that B has occurred is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Conditioning on B restricts attention to the outcomes inside B and rescales their probabilities to sum to 1, which is the role of the denominator.

Theorem 1.16 (Bayes' Theorem). *Bayes' theorem reverses the direction of conditioning, updating a hypothesis as evidence arrives:*

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}.$$

The four pieces are worth memorizing by name:

- $P(A | B)$ is the posterior, the updated belief in A after seeing B
- $P(B | A)$ is the likelihood, how well A explains the evidence
- $P(A)$ is the prior, the belief in A before the evidence
- $P(B)$ is the marginal, the total probability of the evidence

Odds form skips the marginal

Dividing Bayes' theorem for two competing hypotheses cancels the shared marginal $P(B)$:

$$\underbrace{\frac{P(A | B)}{P(A^c | B)}}_{\text{posterior odds}} = \underbrace{\frac{P(B | A)}{P(B | A^c)}}_{\text{likelihood ratio}} \times \underbrace{\frac{P(A)}{P(A^c)}}_{\text{prior odds}}.$$

Each piece of evidence multiplies the prior odds by its likelihood ratio, so repeated independent evidence just multiplies again. This avoids computing $P(B)$ and is the fastest route through two-hypothesis update problems.

Theorem 1.17 (Law of Total Probability). *If $B_1, B_2, \dots, B_n$ partition the sample space (mutually exclusive and exhaustive), then*

$$P(A) = \sum_i P(A | B_i) P(B_i).$$

This is how the marginal in Bayes' theorem is usually assembled. When $P(A)$ is not known directly but the conditional probabilities on each case are, summing those conditionals weighted by the case probabilities builds it. The two-way form $P(A) = P(A | C)P(C) + P(A | C^c)P(C^c)$ covers most interview uses.

Example 1.18 (Monty Hall Problem). Assume the contestant chooses Door 1 and Monty opens Door 2 to reveal a goat. Let A_i be the event the car is behind Door i (prior $P(A_i) = \frac{1}{3}$), and B be the event Monty opens Door 2.

The likelihoods $P(B|A_i)$ are:

- $P(B|A_1) = \frac{1}{2}$ (Monty randomly picks Door 2 or 3)
- $P(B|A_2) = 0$ (Monty cannot reveal the car)
- $P(B|A_3) = 1$ (Monty is forced to open Door 2)

The marginal probability of B is:

$$P(B) = \sum_{i=1}^3 P(B|A_i)P(A_i) = \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) + (0) \left(\frac{1}{3}\right) + (1) \left(\frac{1}{3}\right) = \frac{1}{2}$$

By Bayes' Theorem, the posterior probabilities of winning by staying (A_1) or switching (A_3) are:

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B)} = \frac{\left(\frac{1}{2}\right) \left(\frac{1}{3}\right)}{\frac{1}{2}} = \frac{1/6}{1/2} = \frac{1}{3}$$
$$P(A_3|B) = \frac{P(B|A_3)P(A_3)}{P(B)} = \frac{(1) \left(\frac{1}{3}\right)}{\frac{1}{2}} = \frac{1/3}{1/2} = \frac{2}{3}$$

Switching doubles the probability of winning.

1.4.1 Select Worked Problems

Problem 1.28

I roll two dice. Given that the sum is 8, what's the probability that one of the dice shows a 3?

Solution. The possible ways to roll a sum of 8 are: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2). Of these, two options have a 3 in them, so the answer is $\frac{2}{5} = 0.4$.

Problem 1.29

A bag has 3 red and 2 blue balls. You draw one, don't replace it, then draw another. What's $P(\text{2nd is red} \mid \text{1st was blue})$?

Solution. Once we drew one blue on, the remaining bag composition is 3 red balls and 1 blue one. Then, there is a $\frac{3}{4}$ probability that the second will be red.

Problem 1.30

60% of emails are spam. A filter catches 95% of spam but also flags 3% of legitimate emails. An email is flagged. What's the probability it's actually spam?

Solution. We want $P(\text{Spam} \mid \text{Flagged})$. The likelihood $P(\text{Flagged} \mid \text{Spam})$ is given as 0.95, the prior $P(\text{Spam})$ is 0.6, and the marginal probability $P(\text{Flagged})$ is $0.95 \cdot 0.6 + 0.03 \cdot 0.4 = 0.582$.

Then the posterior is

$$P(\text{Spam} \mid \text{Flagged}) = \frac{P(\text{Flagged} \mid \text{Spam}) \times P(\text{Spam})}{P(\text{Flagged})} = \frac{0.95 \times 0.6}{0.582} \approx 0.979$$

Problem 1.31

You have 2 coins. Coin A is fair (50% heads). Coin B is biased (75% heads). You pick a coin at random and flip it. It lands heads. What's the probability you picked coin B?

Solution. We are looking for the posterior, $P(B \mid H)$.

The priors are $P(A) = P(B) = 0.5$. The likelihoods are $P(H \mid A) = 0.5$ and $P(H \mid B) = 0.75$. Then $P(H) = P(H \mid A) \cdot P(A) + P(H \mid B) \cdot P(B) = 0.5 \cdot 0.5 + 0.75 \cdot 0.5 = 0.625$. Then finally

$$P(B \mid H) = \frac{P(H \mid B) \cdot P(B)}{P(H)} = \frac{0.75 \cdot 0.5}{0.625} = 0.6$$

After flipping and landing on heads, our belief that we picked coin B increased from 0.5 to 0.6.

Problem 1.32

Same as 1.31. The chosen coin is flipped twice and get heads both times. Now what's $P(B)$?

Solution. The only difference is our priors are $P(A) = 0.4$ and $P(B) = 0.6$. Then we update $P(H) = P(H \mid A) \cdot P(A) + P(H \mid B) \cdot P(B) = 0.5 \cdot 0.4 + 0.75 \cdot 0.6 = 0.65$. Finally:

$$P(B \mid H) = \frac{P(H \mid B) \cdot P(B)}{P(H)} = \frac{0.75 \cdot 0.6}{0.65} = \frac{9}{13} \approx 0.692$$

Problem 1.33

A family has 2 children. Given that at least one child is a boy, what's the probability both are boys?

Solution. We can answer this by considering the restriction over the sample space caused by the new information. Originally, for two children there would be four possibilities, of which 1 has two boys. After we know that at least one child is a boy, only three options remain $(B, G), (B, B), (G, B)$. Thus, the probability both are boys is $\frac{1}{3}$.

Problem 1.34

Three cards: one is red on both sides, one is blue on both sides, one is red on one side and blue on the other. You pick a card at random and see a red side. What's the probability the other side is also red?

Solution. The natural first instinct is to treat the cards as the sample space, as with the births problem: restrict to cards consistent with seeing a red face, then count outcomes among those. This fails here because a card is not a single unit of observation. When a card has two sides of the same color, "seeing red" can happen in more than one way for that card. There's no such internal multiplicity in the births problem.

So take the six sides, not the three cards, as the equally likely outcomes. Three of these six are red: two belong to the RR card, one belongs to the RB card. Since each red side was equally likely to be the one we observed, and two of the three red sides have a red back:

$$P(\text{other side red} \mid \text{observed red}) = \frac{2}{3}.$$

Problem 1.35

You have a 6-sided die. You don't know if it's fair or loaded. A fair die shows each number with probability $1/6$. The loaded die shows 6 with probability $1/2$ and each other number with probability $1/10$. You pick a die at random (50-50) and roll a 6. What's $P(\text{loaded})$?

Solution. By Bayes' theorem, with $P(\text{loaded}) = P(\text{fair}) = 0.5$:

$$P(\text{loaded} \mid 6) = \frac{P(6 \mid \text{loaded}) P(\text{loaded})}{P(6)} = \frac{0.5 \times 0.5}{0.5 \times 0.5 + \frac{1}{6} \times 0.5} = \frac{0.25}{1/3} = 0.75$$

Alternatively, work in odds instead of probabilities. Dividing the Bayes' theorem expressions for $P(\text{loaded} \mid 6)$ and $P(\text{fair} \mid 6)$ cancels $P(6)$ from both, leaving:

$$\underbrace{\frac{P(\text{loaded} \mid 6)}{P(\text{fair} \mid 6)}}_{\text{posterior odds}} = \underbrace{\frac{P(6 \mid \text{loaded})}{P(6 \mid \text{fair})}}_{\text{likelihood ratio}} \times \underbrace{\frac{P(\text{loaded})}{P(\text{fair})}}_{\text{prior odds}}$$

So the posterior odds are just the prior odds scaled by the likelihood ratio, no need to compute $P(6)$ at all. Here, the prior odds are $\frac{P(\text{loaded})}{P(\text{fair})} = \frac{0.5}{0.5} = 1$, and the likelihood ratio is $\frac{P(6 \mid \text{loaded})}{P(6 \mid \text{fair})} = \frac{0.5}{1/6} = 3$. So the posterior odds are $1 \times 3 = 3$, i.e. $3 : 1$ in favor of loaded, giving $P(\text{loaded} \mid 6) = 3/4 = 0.75$.

In short, a six is three times more likely to appear if the die is loaded, so each 6 observed multiplies the odds by 3. If we observed *another* 6, the odds would update again: $3 : 1 \rightarrow 9 : 1$, giving $P(\text{loaded}) = \frac{9}{10}$.

1.5 Distributions

A named distribution is a compressed description of a random process. What makes the catalogue useful is not the formulas in isolation but the mapping from a verbal cue in the problem statement to a process, and from that process to a mean, a variance, and a tail. The organizing question for any problem is therefore which process generated the randomness rather than which formula to apply.

Definition 1.19 (Random variable and distribution). A random variable X is a function assigning a number to each outcome in the sample space Ω . Its distribution specifies how probability is spread across the values X can take, and it is described by one of three equivalent objects:

- the probability mass function (PMF) $P(X = x)$ in the discrete case, with $\sum_x P(X = x) = 1$
- the probability density function (PDF) $f(x)$ in the continuous case, with $\int_{-\infty}^{\infty} f(x) dx = 1$
- the cumulative distribution function (CDF) $F(x) = P(X \leq x)$, defined in both cases

The set of values with nonzero probability is the support.

In the continuous case $P(X = x) = 0$ for every single point, so density is not probability. Only integrals of the density over an interval carry probability, which is why continuous questions are answered through the CDF as $P(a < X \leq b) = F(b) - F(a)$.

Recognize the process first

A named distribution (Bernoulli, Binomial, Geometric, Poisson, and so on) is a family arising from a common underlying random process, indexed by one or more parameters such as n and p for the Binomial. Once a problem is matched to its process, the PMF or PDF, mean, and variance follow immediately from the known formulas for the family. Three cues carry most of the discrete cases:

- a *fixed* number of trials with a counted number of successes points to the Binomial
- a *fixed* number of successes with a counted number of trials points to the Geometric or Negative Binomial
- a count over a window of time or space with no natural trial count points to the Poisson

Definition 1.20 (Moment). The n -th moment of a random variable X is $\mathbb{E}[X^n]$, and its n -th central moment is $\mathbb{E}[(X - \mu)^n]$, taken about the mean $\mu = \mathbb{E}[X]$. A moment exists only when the defining expectation converges absolutely, meaning $\mathbb{E}[|X|^n] < \infty$.

Moments are summary numbers describing the shape of a distribution, each one weighting outcomes by a higher power and so responding more strongly to values far from the center. The first four carry standard names and interpretations:

- the first moment $\mathbb{E}[X]$ is the mean, locating the distribution
- the second central moment $\mathbb{E}[(X - \mu)^2]$ is the variance, measuring spread
- the third standardized moment is the skewness, measuring asymmetry
- the fourth standardized moment is the kurtosis, measuring tail weight

Standardizing divides the central moment by σ^n , which makes skewness and kurtosis dimensionless and therefore comparable across distributions. A distribution is not guaranteed to possess any given moment, since sufficiently heavy tails make the integral diverge, and moments fail from the top down in the sense that a finite n -th moment forces every lower moment to be finite as well.

1.5.1 Bernoulli Distribution

Definition 1.21 (Bernoulli). A random variable X follows a Bernoulli distribution with parameter $p \in [0, 1]$, written $X \sim \text{Bernoulli}(p)$, if it takes exactly two values:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

It models a single trial with two outcomes, conventionally labeled success (1) and failure (0).

$$\mu = p \quad \sigma^2 = p(1 - p)$$

Both moments follow in one line. Since X only takes the values 0 and 1, the identity $X^2 = X$ holds pointwise, so $\mathbb{E}[X^2] = \mathbb{E}[X] = p$ and

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1 - p).$$

The variance is largest at $p = \frac{1}{2}$, where the outcome is most uncertain, and shrinks to zero as p approaches 0 or 1, where the result becomes nearly certain.

Building block for other distributions

The Bernoulli is rarely the distribution of interest on its own. It is the atomic unit that other named distributions are built from. A Binomial random variable is a sum of n i.i.d. $\text{Bernoulli}(p)$ trials, and a Geometric random variable counts trials until the first Bernoulli success. Recognizing a Bernoulli inside a larger problem, often through an indicator variable $\mathbf{1}\{\cdot\}$, is frequently the first step in applying linearity of expectation to more complex sums.

1.5.2 Binomial Distribution

Definition 1.22 (Binomial). $X \sim \text{Binomial}(n, p)$ counts the number of successes in n independent trials, each succeeding with the same probability p . Its PMF is

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Proof. Fix one particular sequence of outcomes with k successes and $n - k$ failures. Independence makes its probability the product $p^k (1 - p)^{n-k}$, and that value depends only on the counts, not on the positions. The number of distinct sequences with k successes is the number of ways to choose which trials succeed, namely $\binom{n}{k}$. Summing the common probability over those disjoint sequences gives the stated PMF. $\square$

$$\mu = np \quad \sigma^2 = np(1 - p)$$

Writing $X = \sum_{i=1}^n X_i$ as a sum of i.i.d. $\text{Bernoulli}(p)$ variables gives both moments without touching the PMF. Linearity of expectation yields $\mathbb{E}[X] = np$ with no independence needed. Variance requires independence, which zeroes every covariance cross term and leaves $\text{Var}(X) = n \cdot p(1 - p)$.

Sampling with replacement against without

The Binomial requires that p stay constant across trials, which holds when sampling with replacement. Sampling without replacement changes the composition of the pool after each draw and produces the Hypergeometric distribution instead. When the population is large relative to the sample, the shift in p is negligible and the Binomial is the standard approximation.

Definition 1.23 (Hypergeometric). Drawing n items without replacement from a population of N containing K successes, the number of successes drawn has PMF

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

Its mean is $\mu = n \frac{K}{N}$, matching the Binomial with $p = K/N$, and its variance is

$$\sigma^2 = np(1-p) \frac{N-n}{N-1},$$

where the factor $\frac{N-n}{N-1}$ is the finite population correction.

The mean is unchanged because linearity of expectation ignores the dependence between draws. The variance is lower because the draws are negatively correlated: each draw removes an item from the pool, so a success on one draw leaves fewer successes available and makes a success on the next slightly less likely. That downward pull is exactly the correction factor, which is below 1, so sampling without replacement is strictly less variable than sampling with replacement. It collapses to 1 as N grows relative to n , which recovers the Binomial.

1.5.3 Geometric Distribution

Definition 1.24 (Geometric). $X \sim \text{Geometric}(p)$ counts the number of independent trials up to and including the first success, where each trial succeeds with probability p . Its PMF is

$$P(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

The PMF is a direct product. Reaching the first success on trial k requires $k-1$ failures followed by one success, and independence multiplies those probabilities. The tail probability is equally compact, since needing more than k trials means the first k all failed:

$$P(X > k) = (1-p)^k.$$

$$\mu = \frac{1}{p} \qquad \sigma^2 = \frac{1-p}{p^2}$$

Two conventions

Some texts define the Geometric as the number of *failures before* the first success, with support $k = 0, 1, 2, \dots$ and PMF $(1-p)^k p$. That version has mean $\frac{1-p}{p}$, exactly one less, while the variance $\frac{1-p}{p^2}$ is identical since shifting a variable by a constant does not change its spread. The two differ by the single trial on which the success occurs, so the convention in force must be established before any numerical comparison. The trials convention with mean $\frac{1}{p}$ is the one used throughout these notes.

Theorem 1.25 (Memorylessness). *The Geometric is the only discrete distribution with the memoryless property:*

$$P(X > m + n \mid X > m) = P(X > n).$$

Past failures carry no information about how many further trials are required.

Proof. Apply the definition of conditional probability. The event $\{X > m + n\}$ is contained in $\{X > m\}$, so their intersection is the former:

$$P(X > m + n \mid X > m) = \frac{P(X > m + n)}{P(X > m)} = \frac{(1 - p)^{m+n}}{(1 - p)^m} = (1 - p)^n = P(X > n). \quad \square$$

Memorylessness is the reason the recursive expectation argument works. A coin that has landed tails ten times in a row still needs $\frac{1}{p}$ further flips in expectation, so conditioning on the first trial reproduces the original problem and gives $\mathbb{E}[X] = 1 + (1 - p)\mathbb{E}[X]$, which solves to $\frac{1}{p}$.

1.5.4 Negative Binomial Distribution

Definition 1.26 (Negative Binomial). $X \sim \text{NegBin}(r, p)$ counts the number of independent trials up to and including the r -th success. Its PMF is

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, \dots$$

The binomial coefficient counts the arrangements of the first $r-1$ successes among the first $k-1$ trials. Trial k is forced to be a success, which is what pins the r -th success to the final position rather than anywhere in the sequence.

$$\mu = \frac{r}{p} \quad \sigma^2 = \frac{r(1-p)}{p^2}$$

Both follow from writing $X = G_1 + G_2 + \dots + G_r$, a sum of r independent $\text{Geometric}(p)$ waiting times. Memorylessness is what guarantees each waiting period restarts fresh after a success, so the G_i are i.i.d. The mean is r times the Geometric mean by linearity of expectation. The variance is r times the Geometric variance by *independence*, not by linearity, since variances add only when covariance terms vanish.

1.5.5 Poisson Distribution

Definition 1.27 (Poisson). $X \sim \text{Poisson}(\lambda)$ counts discrete events occurring in a fixed window of time or space, when events arrive independently at a constant average rate λ per window. Its PMF is

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

$$\mu = \lambda \quad \sigma^2 = \lambda$$

The equality of mean and variance serves as a diagnostic. Count data whose sample variance materially exceeds its mean is overdispersed and is not Poisson. This occurs routinely in market data, where order arrivals cluster rather than arriving at a constant rate, and it motivates the mixture and doubly stochastic models that generalize the Poisson.

Theorem 1.28 (Poisson limit of the Binomial). *Fix $\lambda > 0$ and let $n \rightarrow \infty$ with $p = \lambda/n$, so that the expected count $np = \lambda$ stays constant. Then*

$$\binom{n}{k} p^k (1-p)^{n-k} \rightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

Proof. Substitute $p = \lambda/n$ and split the expression into three factors:

$$\underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\rightarrow 1} \cdot \frac{\lambda^k}{k!} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\rightarrow e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\rightarrow 1}.$$

The first factor is a ratio of k terms each approaching n , the third is the standard limit defining $e^{-\lambda}$, and the fourth has a fixed exponent with a base approaching 1. The surviving product is the Poisson PMF. $\square$

The interpretation matters more than the algebra. The Poisson is the law of rare events, meaning many opportunities for an event to occur with a small probability at each one. This is why it models message counts, quote updates, and defaults, where no natural trial count exists but a rate does.

Superposition and thinning

Superposition. Independent Poisson streams merge into a Poisson stream with the summed rate, so $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ independent give $X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$. Combining order flow across venues is the standard application.

Thinning. If each event in a $\text{Poisson}(\lambda)$ stream is independently retained with probability q , the retained events form a $\text{Poisson}(\lambda q)$ stream, and the discarded ones form an independent $\text{Poisson}(\lambda(1-q))$ stream. Filtering a stream to buy orders only is the standard application.

1.5.6 Exponential Distribution

Definition 1.29 (Exponential). $X \sim \text{Exponential}(\lambda)$ is the continuous waiting time until the next event in a Poisson process of rate λ . Its density and CDF are

$$f(x) = \lambda e^{-\lambda x}, \quad F(x) = 1 - e^{-\lambda x}, \quad x \geq 0.$$

$$\mu = \frac{1}{\lambda} \quad \sigma^2 = \frac{1}{\lambda^2}$$

The Exponential is the continuous counterpart of the Geometric, and the pairing runs deep. The Geometric counts discrete trials until a success while the Exponential measures continuous time until an arrival, and each is the unique memoryless distribution on its domain:

$$P(X > s + t \mid X > s) = P(X > t).$$

The proof is the same cancellation as before, using the tail $P(X > x) = e^{-\lambda x}$. The consequence is that the expected remaining wait is $\frac{1}{\lambda}$ regardless of how long the wait has already lasted, so elapsed time carries no information about the time still to come.

Counts against waiting times

Poisson and Exponential describe one process from two directions. If events arrive at rate λ , the *count* in a unit window is $\text{Poisson}(\lambda)$ and the *gap* between consecutive events is $\text{Exponential}(\lambda)$. The identity linking them is that zero arrivals in $[0, t]$ is the same event as a wait longer than t :

$$P(N_t = 0) = e^{-\lambda t} = P(X > t).$$

Minimum of exponentials. If $X_i \sim \text{Exponential}(\lambda_i)$ independently, then $\min_i X_i \sim \text{Exponential}(\sum_i \lambda_i)$, and the probability that stream j arrives first is $\lambda_j / \sum_i \lambda_i$. This is the central result for competing arrivals, such as determining which of several order flows reaches the book first.

1.5.7 Uniform Distribution

Definition 1.30 (Continuous Uniform). $X \sim \text{Uniform}(a, b)$ places equal density across an interval:

$$f(x) = \frac{1}{b-a}, \quad F(x) = \frac{x-a}{b-a}, \quad a \leq x \leq b.$$

$$\mu = \frac{a+b}{2} \quad \sigma^2 = \frac{(b-a)^2}{12}$$

Probability for a continuous Uniform is proportional to length, so $P(c < X < d) = \frac{d-c}{b-a}$ for any subinterval. The variance depends only on the width $b-a$ and not on the location, which is the general behavior of variance under a shift.

Do not mix the discrete and continuous forms

The discrete Uniform on $\{1, 2, \dots, n\}$ has mean $\frac{n+1}{2}$ and variance $\frac{n^2-1}{12}$, which is not the continuous formula with $a = 1$ and $b = n$. A fair die has $\mu = 3.5$ and $\sigma^2 = \frac{35}{12} \approx 2.92$. Substituting into $\frac{(b-a)^2}{12}$ would give $\frac{25}{12}$, which is wrong.

Universality of the uniform

For any continuous random variable X with CDF F , the transformed variable $F(X)$ is $\text{Uniform}(0, 1)$. Running this backwards, $F^{-1}(U)$ with $U \sim \text{Uniform}(0, 1)$ has distribution F . This inverse transform is the basis for generating arbitrary distributions from a uniform random number generator, which is the foundation of Monte Carlo simulation. For the Exponential it gives the explicit sampler $X = -\frac{1}{\lambda} \ln(U)$.

1.5.8 Normal (Gaussian) Distribution

Definition 1.31 (Normal). $X \sim \mathcal{N}(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The two parameters are exactly its mean and its variance, so the distribution is fully specified by its first two moments.

Two structural properties account for most applications.

Standardization. Any Normal reduces to the standard Normal through $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1)$, which is why a single table of Z values suffices for every Normal question.

Closure under addition. A sum of independent Normals is Normal, with means and variances adding:

$$X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2) \implies X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Standard deviations do not add. This is the source of the $\sqrt{T}$ scaling of volatility, since variance grows linearly in time while volatility grows with the square root.

Standard tail masses

Range	Mass inside	Mass outside
$\mu \pm 1\sigma$	68.3%	≈ 1 in 3
$\mu \pm 1.645\sigma$	90%	5% per tail
$\mu \pm 1.96\sigma$	95%	2.5% per tail
$\mu \pm 2\sigma$	95.4%	≈ 1 in 22
$\mu \pm 2.576\sigma$	99%	0.5% per tail
$\mu \pm 3\sigma$	99.7%	≈ 1 in 370

The one-tailed values 1.645 and 2.326 correspond to the 95th and 99th percentiles and are the multipliers underlying standard value-at-risk calculations.

Remark (Why the Normal is the default). The Normal is the maximum entropy distribution among all distributions on the real line with a given mean and variance. Constraining only the first two moments and otherwise assuming as little as possible produces the Normal, which is the formal version of the claim that it encodes minimal structural assumptions. The Central Limit Theorem gives the second, more operational reason for its ubiquity.

Theorem 1.32 (Central Limit Theorem). *Let $X_1, \dots, X_n$ be i.i.d. with finite mean μ and finite variance σ^2 . Then the standardized sample mean converges in distribution to a standard Normal:*

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

The standardization is essential to the statement. The sample mean $\bar{X}_n$ on its own converges to the constant μ by the law of large numbers, and its variance σ^2/n shrinks to zero, so it is the rescaled deviation that stabilizes into a Normal shape rather than the mean itself.

Where the CLT fails

The common rule that $n \geq 30$ suffices is a heuristic, not a theorem, and it breaks in three ways that matter for trading:

- **Finite variance is required.** Heavy-tailed distributions with infinite variance, such as the Cauchy, never converge. The sample mean of Cauchy draws has the same Cauchy distribution as a single draw, so averaging buys nothing.
- **Skewness slows convergence.** Strongly asymmetric or rare-event distributions may need n in the thousands before the Normal approximation is usable in the tails, which is exactly where risk questions live.
- **Independence is required.** Financial returns are volatility clustered and serially dependent, so effective sample size is far below nominal sample size.

Empirical asset returns are leptokurtic, meaning excess kurtosis above the Normal value of 3, so extreme moves occur far more often than a Gaussian model predicts.

1.5.9 Lognormal Distribution

Definition 1.33 (Lognormal). X is lognormal when $\ln X \sim \mathcal{N}(\mu, \sigma^2)$, so $X = e^Y$ for a Normal Y . Its support is $x > 0$, and

$$\mathbb{E}[X] = e^{\mu + \sigma^2/2}, \quad \text{median}(X) = e^\mu, \quad \text{Var}(X) = (e^{\sigma^2} - 1) e^{2\mu + \sigma^2}.$$

The lognormal is the standard model for asset prices, for two structural reasons. Prices are bounded below by zero while a Normal would assign positive probability to negative prices, and returns compound multiplicatively, so the logarithm of the price is a sum of log returns and therefore Normal by the same CLT argument.

The mean sits above the median because the distribution is right-skewed. Exponentiating a Normal stretches the upper values far more than the lower ones, producing a long right tail. The median only tracks the middle draw, which is e^μ , but the mean is dragged upward by that tail to $e^{\mu + \sigma^2/2}$. The two differ by the factor $e^{\sigma^2/2}$, which sits at 1 when there is no variability and grows as the volatility σ increases. More spread in the log means a heavier right tail and a wider gap between the average outcome and the typical one.

This gap is a concrete case of a general rule: averaging after applying a curved function is not the same as applying the function to the average. Here the function is e^x , which curves upward, so $\mathbb{E}[e^Y]$ comes out larger than $e^{\mathbb{E}[Y]}$. The excess is exactly the $e^{\sigma^2/2}$ term. The same correction reappears in the drift of geometric Brownian motion, the standard continuous-time price model, for the same reason.

1.5.10 Reference Tables

Discrete distributions

Distribution	Cue in the problem	PMF	Mean	Variance	Support
Bernoulli(p)	one trial, success or failure	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$\{0, 1\}$
Binomial(n, p)	successes in n fixed trials	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$	$0, \dots, n$
Hypergeom.	successes, no replacement	$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$	np	$np(1-p) \frac{N-n}{N-1}$	$0, \dots, n$
Geometric(p)	trials until 1st success	$(1-p)^{k-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$1, 2, \dots$
NegBin(r, p)	trials until r -th success	$\binom{k-1}{r-1} p^r (1-p)^{k-r}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$r, r+1, \dots$
Poisson(λ)	count at a rate, no trial count	$e^{-\lambda} \frac{\lambda^k}{k!}$	λ	λ	$0, 1, 2, \dots$
Uniform $\{1, n\}$	equally likely integers	$\frac{1}{n}$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$	$1, \dots, n$

Continuous distributions

Distribution	Cue in the problem	PDF	Mean	Variance	Support
Uniform(a, b)	equally likely over a range	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$[a, b]$
Exponential(λ)	wait until next arrival	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$x \geq 0$
Normal(μ, σ^2)	sum of many small effects	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	$\mathbb{R}$
Lognormal(μ, σ^2)	multiplicative growth	$\frac{1}{x\sqrt{2\pi\sigma^2}} e^{-(\ln x - \mu)^2/2\sigma^2}$	$e^{\mu + \sigma^2/2}$	$(e^{\sigma^2} - 1) e^{2\mu + \sigma^2}$	$x > 0$

Moment generating functions

The MGF is $M_X(t) = \mathbb{E}[e^{tX}]$. It determines the distribution uniquely, generates moments through $\mathbb{E}[X^n] = M_X^{(n)}(0)$, and multiplies across independent sums, which is the fastest proof that Binomials and Poissons and Normals are each closed under addition.

Distribution	$M_X(t)$	Distribution	$M_X(t)$
Bernoulli(p)	$1 - p + pe^t$	Poisson(λ)	$e^{\lambda(e^t-1)}$
Binomial(n, p)	$(1 - p + pe^t)^n$	Exponential(λ)	$\frac{\lambda}{\lambda - t}, t < \lambda$
Geometric(p)	$\frac{pe^t}{1 - (1 - p)e^t}$	Normal(μ, σ^2)	$e^{\mu t + \sigma^2 t^2 / 2}$
NegBin(r, p)	$\left(\frac{pe^t}{1 - (1 - p)e^t} \right)^r$	Lognormal	undefined for $t > 0$

As a note on the lognormal entry, the right tail is heavy enough that $\mathbb{E}[e^{tX}]$ diverges for every $t > 0$, which is a compact example of a distribution that has all moments yet no MGF.

How the families connect

From	Relationship	To
Bernoulli	sum of n i.i.d. trials	Binomial
Bernoulli	count trials to 1st success	Geometric
Geometric	sum of r i.i.d. waits	Negative Binomial
Binomial	$n \rightarrow \infty, p \rightarrow 0, np = \lambda$ fixed	Poisson
Binomial	n large, $np(1 - p)$ large	Normal
Poisson	gap between arrivals	Exponential
Geometric	continuous time limit	Exponential
Exponential	minimum of independent draws	Exponential (rates add)
Normal	exponentiate	Lognormal
Any distribution	standardized mean, finite variance	Normal (CLT)
Any continuous	apply own CDF	Uniform(0, 1)

Memorylessness is the thread through the waiting-time column. The Geometric and the Exponential are the unique discrete and continuous distributions with that property, which is why both admit one-line recursive expectation arguments.

1.5.11 Select Worked Problems

Problem 1.36

A basketball player makes 70% of free throws. In 10 attempts, what's $P(\text{exactly 8 makes})$?

Solution. We are given a target out of 10 independent trials, each with the same probability, so we can use the Binomial distribution:

$$P(X = 8) = \binom{10}{8} \cdot 0.7^8 \cdot 0.3^2 \approx 0.233$$

Problem 1.37

A website gets 2 signups per hour on average. What's $P(0 \text{ signups in the next hour})$?

Solution. The problem gives a rate over a timeframe, so this uses the Poisson or Exponential, this time we can use either. Working in hours, the rate is $\lambda = 2$ per hour and the window is $t = 1$.

Poisson. Count the signups directly and ask for zero of them:

$$P(X = 0) = e^{-\lambda} \frac{\lambda^0}{0!} = e^{-2} \approx 0.135.$$

Exponential. Zero signups in the hour is the same event as the wait for the first signup exceeding one hour, which is the tail probability $P(X > t) = 1 - F(t)$:

$$P(X > 1) = 1 - (1 - e^{-\lambda t}) = e^{-\lambda t} = e^{-2} \approx 0.135.$$

The two agree, which is the identity $P(N_t = 0) = P(X > t)$ between a Poisson count of zero and an Exponential wait past t .

Problem 1.38

You keep drawing cards (with replacement) from a deck until you get an ace. Expected number of draws?

Solution. We want to know the expected trials until the first success with replacement, which is the mean of the Geometric distribution:

$$\mathbb{E}[X] = \mu = \frac{1}{p} = \frac{1}{4/52} = 13$$

Problem 1.39

A factory produces items with a 5% defect rate. In a batch of 20, what's the expected number of defective items and $P(\text{at most 1 defective})$?

Solution. We want the successes (defects) in 20 fixed trials, which calls for the Binomial distribution:

$$\mathbb{E}[X] = \mu = np = 20 \cdot 0.05 = 1$$

$$P(X \leq 1) = P(0) + P(1) = \left(\binom{20}{0} \cdot 0.05^0 \cdot 0.95^{20} \right) + \left(\binom{20}{1} \cdot 0.05^1 \cdot 0.95^{19} \right) = 0.358 + 0.377 = 0.736$$

Problem 1.40

On average, a bus arrives every 10 minutes. What's $P(\text{waiting more than 15 minutes})$?

Solution. We want the tail probability of the Exponential distribution, making our unit of time 1 minute, then $\lambda = 0.1$.

$$P(X > 15) = 1 - (1 - e^{-0.1 \cdot 15}) = e^{-1.5} \approx 0.223$$

Problem 1.41

Test scores are Normal with mean 70 and standard deviation 10. What fraction of students score above 90?

Solution. This is simply asking what proportion of a normal distribution exists in the upper right tail two standard deviations from the mean. We know that 95.4% of the data lies within 2σ , then the portion only in the right tail is $\frac{1-0.954}{2} = 0.023 = 2.3\%$

Problem 1.42

You roll a die until you've gotten two 6's total. Expected number of rolls?

Solution. We can use linearity of expectation along with the Geometric distribution (or more directly, the Negative Binomial). Then

$$\mathbb{E}[X] = \frac{r}{p} = \frac{2}{1/6} = 12$$

Problem 1.43

Errors on a page occur at a rate of 0.5 per page (Poisson). A book has 200 pages. What's the expected total errors? What's the approximate probability the book has fewer than 90 errors?

Solution. Scaling the rate to the full book, the total is Poisson with

$$\mathbb{E}[X] = 200 \cdot 0.5 = 100 = \lambda, \quad \sigma = \sqrt{\lambda} = 10.$$

The book total is a sum of 200 independent per-page Poisson counts, so by the CLT it is approximately Normal, and with $\lambda = 100$ well above the $\lambda \approx 10$ threshold the approximation is safe: $X \approx \mathcal{N}(100, 100)$.

Fewer than 90 errors is one standard deviation below the mean, so by the 68% rule

$$P(X < 90) \approx \frac{1 - 0.68}{2} = 0.16.$$

Problem 1.44

You flip a coin with $P(H) = p$. What's the variance of the number of heads in n flips? For what value of p is the variance maximized?

Solution. This is a binary outcome with a fixed number of flips, which calls for the Binomial distribution. The variance is therefore $\sigma^2 = np(1-p)$. This is maximized at $p = \frac{1}{2}$.

1.6 Markov Chains

A Markov chain models a system that moves between states in steps, where the rule for the next move depends only on where the system is now.

Definition 1.34 (Markov Property). A process has the Markov property when the distribution of the next state depends only on the current state. The path taken to reach the current state carries no additional information.

This is memorylessness generalized from waiting times to states. The Geometric and Exponential forgot how long they had already waited. A Markov chain forgets its entire past trajectory and keeps only its current position, which is what makes the current state a sufficient summary for predicting the future.

Definition 1.35 (States and Transitions). A *state* is a complete description of the system at one step. Each state carries *transition probabilities* giving the chance of moving to each other state in one step. A *transient* state is one the chain eventually leaves, and an *absorbing* (terminal) state is one it can never leave, such as “win” or “lose”. Collecting the transition probabilities into a matrix P , with entry P_{ij} the probability of moving from state i to state j , gives the *transition matrix*. Each row sums to 1, since the chain must go somewhere.

Two questions a Markov chain answers

Problems split by whether the chain can get stuck.

- **Absorbing chains** contain terminal states the process eventually lands in. The questions are which absorbing state it reaches, a *hitting probability*, and how long it takes to get there, a *hitting time*. Both are solved by first-step analysis.
- **Recurrent chains** keep moving with no absorbing state. The question is the long-run fraction of time spent in each state, given by the stationary distribution.

1.6.1 Absorbing Chains and First-Step Analysis

A problem with terminal states is solved by **first-step analysis**, which is the same conditioning-on-the-next-move idea behind the recursive expectation examples earlier. Condition on the single step the chain takes from its current state, and write the quantity at that state as a weighted average of the same quantity at the neighboring states, with the transition probabilities as weights. The recipe is:

1. enumerate the states and mark each as transient or absorbing
2. for each transient state, write the quantity of interest as a weighted average over the states reachable in one step
3. use the absorbing states to set the boundary conditions
4. solve the resulting system of equations

The quantity of interest is whichever the problem asks for. Filling it with a probability yields a hitting probability, and filling it with an expected step count yields a hitting time.

Example 1.36 (Gambler's Ruin). A gambler starts with k dollars and bets \$1 at a time, winning a dollar with probability p and losing one with probability $q = 1 - p$. Play continues until the stake reaches \$0 (ruin) or a target $\$N$. The state space is $\mathcal{S} = \{0, 1, \dots, N\}$, with 0 and N absorbing.

Let $u_i = P(\text{ruin} \mid \text{start at } i)$. Conditioning on the first bet gives the first-step equation

$$u_i = p u_{i+1} + q u_{i-1},$$

with boundary conditions set by the absorbing states,

$$u_0 = 1 \quad (\text{already ruined}), \quad u_N = 0 \quad (\text{already at target}).$$

Writing $r = \frac{q}{p}$, the solution for a starting stake k is

$$u_k = \frac{r^k - r^N}{1 - r^N}.$$

In the fair case $p = q = \frac{1}{2}$ this is undefined as written, and the equation instead solves to

$$u_k = \frac{N - k}{N},$$

so the probability of reaching the target is $\frac{k}{N}$, simply the stake as a fraction of the goal.

Expected time to absorption

The expected number of steps until the chain is absorbed, starting from state k , is another first-step analysis. Writing D_k for that expected duration,

$$D_k = 1 + p D_{k+1} + q D_{k-1}, \quad D_0 = D_N = 0,$$

where the leading 1 counts the current step. For the fair Gambler's Ruin ($p = q = \frac{1}{2}$) this solves to

$$D_k = k(N - k).$$

The product form matches intuition. Duration is longest when the gambler starts in the middle, far from both barriers, and shrinks to zero at either edge.

1.6.2 Recurrent Chains and the Stationary Distribution

When a chain has no absorbing states and keeps moving indefinitely, the relevant question is not where it ends but how its time is divided among the states over the long run.

Definition 1.37 (Stationary Distribution). For a chain with transition matrix P , a stationary distribution is a probability vector π satisfying

$$\pi P = \pi, \quad \sum_i \pi_i = 1.$$

It is the long-run fraction of time the chain spends in each state. The equation $\pi P = \pi$ says that a chain already distributed as π reproduces that same distribution after one step.

For a chain that is irreducible and aperiodic, meaning every state is reachable from every other and the chain does not cycle in a fixed period, this π is unique and the chain settles into it regardless of where it started.

1.6.3 Select Worked Problems

Problem 1.45

A gambler has \$4 and wants to reach \$7, betting \$1 each round on a fair game. What's the probability of reaching \$7 before going broke?

Solution. Since the game is fair, we can assume that $p = \frac{1}{2}$. Then the probability of reaching the target is $\frac{k}{N} = \frac{4}{7}$

Problem 1.46

You flip a coin. What's the expected number of flips to get 3 heads in a row?

Solution. $\mathcal{S} = \{S_0, S_1, S_2, S_3\}$, where the subscript indicates the number of consecutive heads in the state. Then the only absorbing state is S_3 .

We then create a system of transition equations, where each state has a 50% chance to progress or to go back to zero consecutive heads:

$$E_0 = 1 + \frac{1}{2}E_0 + \frac{1}{2}E_1$$

$$E_1 = 1 + \frac{1}{2}E_0 + \frac{1}{2}E_2$$

$$E_2 = 1 + \frac{1}{2}E_0 + \frac{1}{2}E_3$$

$$E_3 = 0$$

We then backsubstitute to solve for E_0 :

$$E_2 = 1 + \frac{1}{2}E_0 + \frac{1}{2}0 = 1 + \frac{1}{2}E_0$$

$$E_1 = 1 + \frac{1}{2}E_0 + \frac{1}{2}\left(1 + \frac{1}{2}E_0\right) = \frac{3}{2} + \frac{3}{4}E_0$$

$$E_0 = 1 + \frac{1}{2}E_0 + \frac{1}{2}\left(\frac{3}{2} + \frac{3}{4}E_0\right) = \frac{7}{4} + \frac{7}{8}E_0$$

$$\rightarrow E_0 = 14$$

Problem 1.47

You roll a die repeatedly. What's the expected number of rolls until the running sum is a multiple of 3?

Solution. Divisibility by three makes the relevant states the residue classes of the running sum modulo 3: $\mathcal{S} = \{[0], [1], [2]\}$, with $[0]$ the target. A die value is uniform mod 3, since the faces $1, \dots, 6$ give residues $1, 2, 0, 1, 2, 0$, two of each. Adding a uniform residue to any current sum lands on a uniform residue, so each roll moves to one of the three classes with probability $\frac{1}{3}$. This folds the six equally likely rolls into three equally likely transitions.

The state $[0]$ plays two roles that must be separated. As the *target* it is absorbing, so no further rolls are needed once the sum is a multiple of 3, i.e., $E_0 = 0$.

As the *start* the running sum is 0, but the target requires arriving there by a roll, so the first move is forced and the start does not inherit the absorbing value. It gets its own equation below. Let E_i be the expected additional rolls to reach a multiple of 3 from residue i . The transient states [1] and [2] give

$$E_1 = 1 + \frac{1}{3}E_0 + \frac{1}{3}E_1 + \frac{1}{3}E_2$$

$$E_2 = 1 + \frac{1}{3}E_0 + \frac{1}{3}E_1 + \frac{1}{3}E_2$$

By symmetry $E_1 = E_2 = E$, and with $E_0 = 0$ each reduces to

$$E = 1 + \frac{2}{3}E \longrightarrow \frac{1}{3}E = 1 \longrightarrow E = 3$$

Finally the forced first roll from the start:

$$R = 1 + \frac{1}{3} \underbrace{E_0}_{=0} + \frac{1}{3}E_1 + \frac{1}{3}E_2 = 1 + \frac{1}{3}(3) + \frac{1}{3}(3) = 3$$

Checks. Each roll independently lands on a multiple of 3 with probability $\frac{1}{3}$ regardless of the current sum, so the count is Geometric($\frac{1}{3}$) with mean $\frac{1}{p} = 3$. Equivalently, this is the expected return time to state [0], which for an irreducible chain is $\frac{1}{\pi_0}$. The residue chain is symmetric with $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, giving $\frac{1}{\pi_0} = 3$.

Problem 1.48

A particle starts at position 0 on a number line. Each step: +1 with probability $\frac{2}{3}$, -1 with probability $\frac{1}{3}$. What's the probability it ever reaches -3?

Solution. With no upper barrier, this is one-sided ruin: against an upward drift, what is the chance the walk ever gets 3 levels down. For a biased walk that steps toward the target with probability p and away with probability p' , where $p < p'$, the probability of ever reaching a point n levels below the start is $(p/p')^n$. The ratio is below 1 since the walk drifts away from the target, and it is raised to the n because the n levels are cleared one at a time, each an independent copy of the same descent. Here the walk moves down with probability $p = \frac{1}{3}$ and up with probability $p' = \frac{2}{3}$, and $n = 3$, so

$$P(\text{ever reach } -3) = \left(\frac{1/3}{2/3}\right)^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

This ratio is the same $r = q/p$ that drives Gambler's Ruin, here with the upper barrier sent to infinity. When the walk is fair the ratio equals 1 and every level is reached with certainty.

2 Brainteasers

2.1 Strategies

Most of these problems are solved by recognizing a recurring structure rather than by extended calculation. A given problem usually reduces to one of a small set of standard techniques, and the central skill is identifying which one applies. The approach proceeds in three phases: frame the problem precisely, map it to a known reduction, and verify the result against limiting cases.

2.1.1 Framing

Much of the work happens at the setup stage. An error introduced while framing a problem propagates through every later step and cannot be repaired by correct algebra, so the setup warrants explicit attention before any technique is chosen.

Framing checklist

- State every implicit assumption: fairness of coins and dice, independence of trials, sampling with or without replacement, and the exact meaning of “random”, meaning which distribution over which set
- Fix the sample space and the unit of observation. Counting the wrong objects as equally likely is the most common source of probability errors
- Classify the target quantity as a probability, an expectation, a count, a ratio, an optimal strategy, or a worst-case guarantee. The class selects the toolset

The choice of observational unit deserves particular care. A single experiment supports several sample spaces, and only some of them make the outcomes equally likely. A direct count is valid only at the level where that symmetry holds, so the first task is to identify that level and count there.

2.1.2 Core Probabilistic Reductions

Most probability puzzles reduce to one of the techniques below:

Cue to reduction

Cue in the problem	Reduction
“at least one”	complement, $1 - P(\text{none})$
“expected number of ...”	linearity of expectation with indicators
exchangeable items, “ A before B ”	symmetry
“until the first ...”, repeated identical trials	conditioning on the first step
waiting for a pattern, random walk, absorbing states	Markov states
updating on evidence, false positives	Bayes, odds form
fixed trial count, rate, or waiting structure	recognize the named distribution
sum or average of many independent pieces	normal approximation, CLT

Three techniques recur often enough to warrant restating:

Linearity with indicators. The expected size of a count equals the sum of the probabilities of its parts, whether or not those parts are independent. The method is to write the count as a sum of indicator variables, one per event, and take expectations term by term. Its power comes from sidestepping dependence entirely, since linearity holds without any independence assumption, which reduces an intractable joint distribution to a list of individual probabilities.

Symmetry. When several items are exchangeable, every statement about their relative order is uniform over the possible orderings. This converts a probability into a counting ratio with no further work. Two consequences appear repeatedly: a specific item A precedes a specific item B with probability $\frac{1}{2}$, and among k competing items each is equally likely to be first. When the items instead compete at rates λ_i , the uniform $\frac{1}{k}$ generalizes to the rate share $\lambda_j / \sum_i \lambda_i$.

Conditioning and self-similarity. A process is self-similar when its state after one step is a scaled copy of the original problem. Conditioning on that first step then expresses the unknown in terms of itself, producing a solvable equation. This is recursive expectation for waiting times and first-step analysis for Markov chains, and it is the standard route for any “expected time until” quantity.

2.1.3 Combinatorial and Logical Tactics

These tactics address counting problems and questions of the form “is a configuration reachable” or “must something exist”.

Small cases and induction. Compute the answer for $n = 1, 2, 3$, conjecture the pattern, then confirm it by induction or a direct argument. Beyond producing the formula, small cases expose the recursive structure that conditioning later exploits.

Invariants. An invariant is a quantity left unchanged by every legal move. If two states differ in the value of some invariant, no sequence of moves connects them, which resolves reachability and “is it possible” questions without any search. The usual construction assigns each configuration a value in a small set, often through a parity or coloring argument, and shows that every move preserves it.

Pigeonhole and extremal arguments. If more items than categories are distributed among the categories, some category holds at least two. The extremal variant examines the largest or smallest element of a configuration to force a contradiction or an existence claim. Both establish that something must occur without exhibiting it, which is what makes them suited to existence questions.

Working backwards. When the goal state is simple and the starting state is complex, reason from the end. The value of the final position is known outright, and each earlier position is defined in terms of positions closer to the goal, which is the structure underlying optimal-stopping and game problems.

Generating functions. Encoding a sequence as the coefficients of a power series turns operations on the sequence into algebra on the series. Convolution becomes multiplication, so counting problems with additive constraints and distributions of sums of independent variables both reduce to expanding a product.

2.1.4 Adversarial and Sequential Decisions

These problems involve a decision maker choosing actions over time, sometimes against an opponent, to optimize an outcome.

Backward induction. In a finite game of alternating moves with perfect information, solve from the last move outward. At each position a player selects the move that optimizes their outcome assuming optimal play thereafter. Applied recursively, this fixes the value of every position together with the equilibrium strategy.

Dominant strategies and minimax. A dominant strategy is optimal regardless of the opponent’s choice, and when one exists the analysis reduces to it. When none exists, a zero-sum game is solved by minimax, in which each player minimizes their maximum loss. With

mixed strategies permitted, the two players' values coincide at a single value of the game, which is von Neumann's minimax theorem.

Optimal stopping. When observations arrive in sequence and each accept-or-reject decision is final, the problem is one of optimal stopping, and its solution is typically a threshold rule. Observe without committing for a calibrated initial portion of the sequence to establish a baseline, then accept the first later observation that exceeds it. In the canonical case of selecting the best of n candidates presented in random order, the initial portion is a fraction $1/e$ and the resulting success probability also approaches

$$\frac{1}{e} \approx 0.368$$

Fair games and martingales. A fair bet has expected value zero, and a sequence of fair bets forms a martingale, whose expected future value equals its present value regardless of history. No stopping rule applied with finite capital converts a fair game into a favorable one in expectation, which is the formal reason betting systems fail. Gambler's Ruin is the standard model, and its ruin and duration formulas quantify the outcome.

Remark (Kelly sizing). For a favorable repeated bet, the wealth fraction that maximizes long-run growth is the edge divided by the odds, which for an even-money bet at win probability p is $2p - 1$. Staking more than this fraction raises variance without raising growth, and staking well beyond it drives the long-run growth rate negative despite the positive edge.

2.1.5 Counting Information

A separate class of problems asks for the minimum number of measurements that guarantee a result in the worst case. The governing tool is outcome counting. A measurement with b distinguishable results partitions the remaining possibilities into b groups, so k measurements distinguish at most b^k cases. Resolving M possibilities therefore requires

$$k \geq \log_b M$$

Worst-case guarantees by outcome counting

Measurement	Outcomes per step	k steps resolve
yes/no question	2	2^k
balance weighing	3	3^k

A balance produces three outcomes, left heavy, right heavy, or level, so k weighings resolve 3^k cases, and locating one heavy item among N of known direction needs $\lceil \log_3 N \rceil$. If the odd item may be either heavier or lighter, each candidate splits into two hypotheses and the requirement becomes $3^k \geq 2N$.

The same bound gives $\lceil \log_2 M \rceil$ yes/no questions, achieved by binary search. When the possibilities are not equally likely, the bound sharpens from a raw count to the entropy: the expected number of yes/no questions needed is at least $H(X)$ bits.

2.1.6 Estimation

Order-of-magnitude questions are handled by decomposition. Break the unknown into a product of factors, each estimable to within a small multiple, anchor on a known reference quantity, estimate each factor independently, and combine. Because the factor errors are multiplicative and roughly independent, they tend to partially cancel rather than accumulate, so a chain of rough estimates usually lands within a factor of two or three of the true value.

2.1.7 Verification

A closed-form answer should survive a set of inexpensive checks before it is trusted. Each check tests the answer against a case where the correct value is already known or constrained.

Sanity checks

- **Boundary.** Send a parameter to an extreme, such as $p \rightarrow 0$, $p \rightarrow 1$, $n = 1$, or $n \rightarrow \infty$, and confirm the formula reduces to the obvious value
- **Range.** A probability lies in $[0, 1]$, an expected number of trials is at least 1, and a variance is nonnegative
- **Monotonicity.** The answer should move in the correct direction as each parameter increases
- **Symmetry.** If the setup is symmetric in two variables, the answer must be symmetric in them as well
- **Small case.** Evaluate by hand at $n = 2$ or 3 and match the result against the formula

2.2 Select Worked Problems

Problem 2.1 [Easy]

Five pirates need to divide 100 Gold Coins. Pirates have a hierarchy, from Level 1 to level 5. The highest level pirate is the leader. The leader proposes a plan to distribute the gold and all the pirates vote on it (including the leader). If at least 50% of the pirates agree on the plan, the gold is split according to the proposal. If not, level-5 pirate is kicked from the ship, and the level-4 pirate now proposes a new plan. This process continues until a proposal is accepted. All pirates are extremely smart and extremely greedy. How does level-5 Pirate divide the treasure?

Solution. We approach this problem game-theoretically and dynamically:

- 2 pirates → Pirate 2 proposes 100 coins for themselves and zero for Pirate 1
- 3 pirates → Pirate 3 bribes Pirate 1 with one coin and keeps 99 for themselves, which they accept because they get zero coins if Pirate 3 is booted off.
- 4 pirates → Pirate 4 bribes Pirate 2 with one coin and keeps 99 for themselves, in the same manner as the 3-pirate case.
- 5 pirates → Pirate 5 bribes pirates 1 and 3 with one coin each, keeping 98 for themselves.

Thus, the final division is $[98, 0, 1, 0, 1]$.

Problem 2.2 [Easy]

An explorer travels 1 mile South, 1 mile East, then 1 mile North, and finds themselves back at the starting point. The explorer spots a bear, what is the color of the bear?

Solution. The bear must be white. There are a few relevant scenarios that could have resulted in the explorer starting back where they began. In the first case, they started at the North pole, then travelling South equates to travelling directly away from the Pole. They can then travel *any* distance East or West, the following mile back North will return them to their destination. There are no bears in the South Pole so we omit this, though the same logic applies (with the Eastward movement equivalent to spinning in place or not moving at all).

There are other solutions near the South Pole. If the traveler's initial one-mile journey South ends on a latitude circle with a circumference of $\frac{1}{n}$ ($n \in \mathbb{Z}^+$) miles, walking one mile East will complete n full rotations around the pole and leave them exactly where they started walking East, meaning the following mile North would place them in the same starting position.

Problem 2.3 [Easy]

There is a light bulb inside a room and three switches outside. All switches are currently in off state and only one switch controls the light bulb. You may turn any number of switches on or off any number of times you want. How many times do you need to go into the room to figure out which switch controls the light bulb?

Solution. We only need to enter the room once. The first switch is flipped and left on for enough time that the bulb become hot. It is then flipped off and the second switch is flipped, we then immediately enter the room. This leads the three cases:

1. Bulb is hot → the light must be off and the first switch controls the light

2. Bulb is on $\rightarrow$ the second switch controls the light
3. Bulb is off and cold $\rightarrow$ the third switch controls the light

If there were four switches, we can still infer the switch in one entry by turning on both switches 1 and 2 for a long time, then switching off switch 2 and turning on switch 3 before entering the room. This results in the following cases:

1. Bulb is on and hot $\rightarrow$ Switch 1 controls the light
2. Bulb is on and cool $\rightarrow$ Switch 3 controls the light
3. Bulb is off and hot $\rightarrow$ Switch 2 controls the light
4. Bulb is off and cool $\rightarrow$ Switch 4 controls the light

The extension to four bulbs worked because we had two binary variables (illumination and bulb temperature) which creates 4 distinct states, of which we were only using 3 in the 3-bulb case.

Problem 2.4 *[Easy]*

How do you place 50 good candies and 50 rotten candies in two boxes such that if you choose a box at random and take out a candy at random, it better be good!

We need to maximize the probability of getting a good candy when selecting a random box and a random candy from it.

Solution. On one extreme, we can place all of the good candies in one box with all of the bad candies in another. This gives us $\mathbb{E}[G] = 0.5$. Thus, we should find solutions that better than 50%. We can beat this rather simply, by placing one good candy in one box and the other 49 in the other box with all of the bad candies. This results in:

$$\mathbb{E}[G] = 0.5 * \frac{1}{1} + 0.5 * \frac{49}{99} \approx 0.747$$

Problem 2.5 *[Easy]*

In a world where everyone wants a girl child, each family continues having babies till they have a girl. What do you think will the boy-to-girl ratio be eventually?

Solution. The stopping rule shapes each family but does not affect the aggregate ratio, since it cannot bias any individual birth. Each birth is an independent fair coin, and the choice of when to stop having children cannot change the overall fraction of each sex.

Counting round by round makes this precise. Begin with N couples. In the first round all N give birth, producing $\frac{N}{2}$ girls and $\frac{N}{2}$ boys, and the couples with a girl stop. The $\frac{N}{2}$ couples with a boy have another child, producing $\frac{N}{4}$ girls and $\frac{N}{4}$ boys, and the process repeats. Each round is an independent batch of fair births and contributes equal numbers of girls and boys, so the totals remain equal and the ratio is

$$1 : 1$$

The appearance of a girl surplus comes from the family structure. Every family terminates on exactly one girl, which makes girls salient as the final child. The boys are the births preceding that girl, and their count follows a geometric distribution with $p = \frac{1}{2}$ and mean $\frac{1-p}{p} = 1$. The

average family therefore has one boy and one girl. The rule fixes the position of the girl within each family, not the proportion of each sex overall.

Problem 2.6

Hundred tigers and one sheep are put on a magic island that only has grass. Tigers can live on grass, but they want to eat sheep. If a Tiger bites the Sheep then it will become a sheep itself. If 2 tigers attack a sheep, only the first tiger to bite converts into a sheep. Tigers don't mind being a sheep, but they have a risk of getting eaten by another tiger. All tigers are intelligent and want to survive. Will the sheep survive?

Solution. This is a parity argument by backward induction, sharing the alternating logic of the pirate loot-splitting problem. The cases build from the bottom up, and the governing fact is that a tiger eats the sheep only when doing so leaves it safe, meaning the resulting one-sheep island has an outcome in which that new sheep is not itself eaten.

Consider the smallest cases. With one tiger, no other tiger threatens it, so it eats the sheep and survives. With two tigers, a tiger that eats the sheep becomes a sheep facing one remaining tiger, which is the one-tiger case in which the sheep is eaten, so neither tiger risks it and the sheep survives. With three tigers, a tiger that eats reduces the island to the two-tiger case, in which a sheep survives, so eating is safe and the sheep is eaten. With four, eating reduces to the three-tiger case, in which the sheep is eaten, so no tiger risks it and the sheep survives.

Each case is thus determined by the one below it: the sheep is eaten precisely when the case with one fewer tiger leaves the sheep safe. This alternation follows parity, so the sheep survives for an even number of tigers and is eaten for an odd number. With 100 tigers, the sheep survives.

Problem 2.7

A duck is sitting at the center of a circular lake. A fox is waiting at the shore, not able to swim, wishing to eat the duck. The Fox can move around the whole lake at a speed four times the speed at which the duck can swim. The duck can fly, but only once it reaches the shore of the lake, it can't fly from the water directly. Can the duck always reach the shore without being eaten by the fox?

Solution. Let R be the radius of the lake, v the duck's speed, and $4v$ the fox's. A direct dash across fails: the duck would swim R while the fox runs at most half the circumference, πR . Since $\pi < 4$, the fox covers its distance in less time and arrives first.

Instead the duck swims in a small circle concentric with the lake, of radius slightly less than $\frac{R}{4}$. The choice of $\frac{R}{4}$ comes from comparing angular speeds. A duck at radius r sweeps angle at rate $\frac{v}{r}$, while the fox running the shore sweeps at rate $\frac{4v}{R}$, so the duck out-rotates the fox exactly when

$$\frac{v}{r} > \frac{4v}{R} \iff r < \frac{R}{4}$$

The fox tracks the duck from the nearest point of the shore, but since the duck gains angle steadily, the separation between them grows until the duck sits diametrically opposite the fox.

At that moment the duck turns and swims straight for the shore. It must cover $R - \frac{R}{4} = \frac{3R}{4}$, while the fox must run half the circumference, πR . Their times are

$$t_{\text{duck}} = \frac{3R}{4v}, \quad t_{\text{fox}} = \frac{\pi R}{4v}$$

so the comparison reduces to $3 < \pi$, which holds. The duck reaches the shore first and escapes.

Problem 2.8

An airplane flies in a straight line from airport A to airport B , and then back in a straight line from B to A . It travels with a constant engine speed and there is no wind. Will the total travel time for the same round trip be greater, less or the same if, throughout both the flights a constant wind blows from A to B ?

Solution. The round trip takes longer with wind. Take a plane speed of 100 mph, a wind of 10 mph, and a distance of 100 miles each way. The wind adds to the ground speed on one leg and subtracts on the other, giving

$$\frac{100}{110} + \frac{100}{90} \approx 2.02 \text{ hours}$$

against 2 hours with no wind.

In general, for distance d , plane speed v , and wind $w < v$, the round trip takes

$$\frac{d}{v+w} + \frac{d}{v-w} = \frac{2dv}{v^2 - w^2}$$

which exceeds the windless time $\frac{2d}{v}$ for any $w \neq 0$, since $v^2 - w^2 < v^2$. The asymmetry is that the plane spends less time on the fast leg and more time on the slow one, so the time lost to the headwind always outweighs the time gained from the tailwind.

Problem 2.9

You are given two cords that both burn in exactly one hour, not necessarily at a uniform rate. How should you light the cords to determine a time interval of exactly 15 minutes?

Solution. We light three of the four ends at the same time. When the cord with both ends lit burns out, thirty minutes will have passed. At this point we snuff the fire on the other cord. The segment remaining can be lit from both sides and will burn for an interval of 15 minutes.

The same interval can be obtained from a single cord by maintaining four simultaneous flames, assuming that cutting and lighting take negligible time. A cord holds sixty minutes of burn time, and with k flames alight that reserve is consumed at rate k , so four flames exhaust it in $\frac{60}{4} = 15$ minutes

To begin, we cut the cord in two and light all four ends. Whenever a piece burns out and the count drops to two flames, we cut one of the remaining pieces and light the two new ends, restoring the count to four. The number of cuts required is not bounded in advance, since it depends on how unevenly the cord burns, but the flame count is held at four throughout and the cord is spent after exactly fifteen minutes.

Problem 2.10

There are 10 black socks and 10 red socks in a cupboard. Your task is to draw a few socks at random, without looking, such that you get at least one matching pair (single color). What's the least number of socks that you need to draw?

Solution. You only need to draw three. If the first two were opposite colours (not a match), then the third sock *must* match one of the first two. This is the pigeonhole principle with 3 pigeons (socks drawn) and two holes (colours).

Problem 2.11

Prove that there are two opposite points on the Earth's surface that have the same temperature. Assume that the temperature varies continuously over the surface.

Solution. Consider any great circle on the surface, such as the equator, and let $t(\theta)$ be the temperature at angle θ . Two points are opposite when their angles differ by π , so the goal is a value of θ satisfying $t(\theta) = t(\theta + \pi)$. Define $f(\theta) = t(\theta) - t(\theta + \pi)$, which is zero when the two points have equal temperatures. As a difference of continuous functions, f is continuous.

Evaluating at the ends of a half circle, and using $t(2\pi) = t(0)$ because the circle closes on itself,

$$f(0) = t(0) - t(\pi), \quad f(\pi) = t(\pi) - t(0) = -f(0)$$

so f takes values of opposite sign at 0 and π , or is zero, in which case the temperatures are equal and we are done. Otherwise, by the Intermediate Value Theorem there must be a point on the closed interval $[0, \pi]$ such that $f(\theta) = 0$, and that root is an antipodal pair at equal temperature.

This is the one-dimensional case of the Borsuk-Ulam theorem, which states that any continuous map $S^n \rightarrow \mathbb{R}^n$ admits an antipodal pair with equal values. Thematically it resembles the hairy ball theorem, in that a continuous structure on a sphere is forced into a degeneracy, though Borsuk-Ulam constrains values at antipodal pairs while the hairy ball theorem forces a tangent vector field to vanish somewhere.

Problem 2.12

The probability of having at least one accident on a road in one hour is $\frac{3}{4}$. What is the probability that at least one accident occurs in half an hour?

Solution. Assuming accidents in disjoint intervals are independent, no accident in an hour means no accident in each of its two half-hour intervals. Let q be the probability of no accident in half an hour. Then $q^2 = 1 - \frac{3}{4} = \frac{1}{4} \rightarrow q = \frac{1}{2}$. Then the probability of at least one accident in half an hour is $1 - q = \frac{1}{2}$.

Problem 2.13

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. Now, do you want to pick door No. 2? What is the probability to win the car if you switch?

Solution. Switching wins with probability $\frac{2}{3}$.

Suppose door 1 is chosen. The car is equally likely to be behind each door, giving three cases. If the car is behind door 1, the host opens one of the goats and switching loses. If the car is behind door 2, the host is forced to open door 3, and switching wins. If the car is behind door 3, he is forced to open door 2, and switching wins.

Switching therefore wins in two of the three equally likely cases. Equivalently, switching wins exactly when the initial pick was wrong, which occurs with probability $\frac{2}{3}$.

Problem 2.14

One hundred prisoners are lined up, facing one direction and assigned a random hat, either red or blue. Each prisoner can see the hats in front of them but not behind. Starting with the prisoner at the back of the line and moving forward, they must each, in turn, say only one word which must be "red" or "blue". If the word matches their hat color they are released, if not, they are kept imprisoned. They can hear each others' answers, no matter how far they are on the line, but they do not hear the verdict (whether the answer was correct). A friendly guard warns them one night before, giving them enough time to come up with a strategy. How many prisoners can be freed using the best strategy? Assume that there is an unknown number of red & blue hats.

Solution. Ninety-nine prisoners are guaranteed release, and the first to speak is correct with probability $\frac{1}{2}$. This is optimal, since that prisoner sees only the hats ahead and receives no information bearing on their own, so no strategy improves on a coin flip for them.

The prisoners agree in advance to encode parity. The prisoner at the back, who sees all 99 hats ahead, says "red" if the number of red hats ahead is even and "blue" if it is odd.

Each later prisoner deduces their own colour from three known quantities: the announced parity, the colours called behind them, and the hats visible ahead. Their hat is red exactly when the combined parity of the reds heard and seen differs from the announced parity, since their own hat is the only one unaccounted for and so must absorb the discrepancy. Having deduced it, they answer correctly, and their answer joins the record used by those ahead.

Problem 2.15

A traveler meets a pair of twins at a fork in the road: one path is correct and one incorrect. One of the twins always tells the truth (knight), and the other one always lies (knave). Which single question can the traveler ask one of the twins to determine the correct path?

Solution. Ask either twin: "Which path would your brother say is the good one?", then take the opposite of the path indicated.

The question works because it forces the answer to pass through both twins, one truthful and one lying, so a single inversion is applied no matter which twin is asked. Let the honest report of the good path be G . The brother being described contributes one inversion if he is the liar, and the twin answering contributes a second inversion if he is the liar. Exactly one of the two is the liar in either case, so the answer always carries exactly one inversion and points to the bad path.

Concretely, asking the knight gives a truthful report of what the knave would say, and the knave would name the bad path. Asking the knave gives a lying report of what the knight would say, and the knight would name the good path, which the knave inverts to the bad path. Both routes indicate the bad path, so the traveler takes the other one.

Problem 2.16

There are two jars, one contains water, and the other contains wine (equal volumes). A certain amount of water is transferred to the wine, then the same amount of the mixture is transferred back to the water beaker. Is there now more water in the wine than there is wine in the water?

Solution. Call the initial volume in each jar V . Each jar ends with volume V again, since equal amounts move out and back. Say the wine jar finishes with x of water, so its remaining wine is $V - x$.

Since there was initially V wine, the x missing from the wine jar must be in the water jar, with $V - x$ of water there to reach a total volume of V . Thus the amount of water in the wine jar equals the amount of wine in the water jar.

Problem 2.17

There are 100 light bulbs in a sequence, all kept in OFF state. The first person comes and toggles all the bulbs which are at a position that is a multiple of 1, i.e. switch all bulbs to ON. The second person toggles all multiples of 2, i.e. turn the even-numbered bulbs OFF and leave the odd ones as is. The third person comes and toggles all multiples of 3. This process continues until the 100th person does their part. After this, how many bulbs are in ON state?

Solution. 10 bulbs will be on. A bulb ends ON if it is toggled an odd number of times, and bulb n is toggled once by each person whose number divides n , so it ends ON exactly when n has an odd number of divisors. This happens only for perfect squares, and the number of perfect squares up to n is $\lfloor \sqrt{n} \rfloor$, which is $\lfloor \sqrt{100} \rfloor = 10$.

The prime factorization of a number n follows the form $n = p_1^{e_1} \cdot p_2^{e_2} \cdots p_k^{e_k}$, where e_i counts the copies of the prime p_i . A divisor d of n has to use a subset of that factorization: $d = p_1^{a_1} \cdot p_2^{a_2} \cdots p_k^{a_k}$, $0 \leq a_i \leq e_i$. There are $e_i + 1$ choices for each exponent, so the total number of divisors is

$$\tau(n) = \prod_{i=1}^k (e_i + 1)$$

n is a perfect square when every e_i is even, since squaring doubles each exponent. Then each $e_i + 1$ is odd, and a product of odd numbers is odd, so a perfect square has an odd number of divisors and no other number does.

Problem 2.18

We have a weighted coin that shows a Head with probability p and Tails with $1-p$, ($0.5 < p < 1$). How do we define two equally likely events using this coin? Each event is a function of the outcomes of one or more coin tosses.

Solution. We want two equally likely events, each built from coin toss outcomes. We do *not* know p , so we anchor the equality on something independent of p . Flip the coin twice and read the ordered pair. Define Event 1 as HT and Event 2 as TH , and discard HH or TT .

These are equally likely because $P(HT) = p(1-p)$ and $P(TH) = (1-p)p$ are the same product, equal for every p . Discarding the tied outcomes HH and TT leaves the remaining probability split evenly between the two.

The flips must be read in fixed, non-overlapping pairs. If instead we flipped a single stream until either HT or TH first appeared, the result would be decided by the very first flip, since the run ends as soon as the coin differs from where it started. That favors HT with probability p and breaks the fairness.

Problem 2.19

Spiderman has two close friends, Mary Jane & Gwen Stacy. After every mission, he rushes to the central subway. One line heads towards Mary's place, and another towards Stacy. Trains from each line leave every 10 minutes. Spiderman being impartial always boards the first train that leaves. However, he observes that he ends up visiting Mary Jane nine times more often than Gwen Stacy. Can you decipher why?

Solution. The key is that Spiderman boards the next train to depart, so what matters is the waiting gap *before* each train, and those two gaps are unequal. We assume his arrival time is uniform across the 10-minute cycle.

Say Mary's train departs at times 0 and 10, and Gwen's train departs at some time d in between, $0 < d < 10$. If he arrives in $(0, d)$ the next train is Gwen's, and if he arrives in $(d, 10)$ the next train is Mary's. So the interval leading to Gwen has length d and the interval leading to Mary has length $10 - d$, giving $P(\text{Mary}) = \frac{10-d}{10}$ and $P(\text{Gwen}) = \frac{d}{10}$. Concretely, we solve for a 9-to-1 ratio: $\frac{10-d}{d} = 9 \implies d = 1$.

So Gwen's train leaves just 1 minute after Mary's, which leaves a 9-minute window in which the next train goes to Mary.

Problem 2.20

An egg breaks only if dropped from above a threshold floor, within a 100-story building. Every time you drop an egg, it is counted as an attempt. You are given 2 eggs to deduce the threshold floor, and eggs can be reused if they do not break. Minimize the number of drops (trials) in the worst case.

Solution. We cannot perform a standard binary search because overestimates of the target require a linear search from the last floor to the threshold (until we find it). Considering this, we could try to optimize an equation which divides 100 into d splits and then performs $\frac{100}{d}$ trials for the linear scan:

$$\min \left(d + \frac{100}{d} \right), \quad d \in \mathbb{N}, d \in [1, 100]$$

The minimum of this can be solved using the roots of the derivative:

$$\frac{d}{dd} \left(d + \frac{100}{d} \right) = 0 \longrightarrow 1 - \frac{100}{d^2} = 0 \longrightarrow d = 10$$

which gives 19 trials in the worst case (threshold on floor 99). This is not the true optimum, the approach failed for two reasons. Firstly the function is discrete, which we only *approximated* using a derivative. Secondly, and more importantly, the equation we chose encoded a constraint that each division be the same size. To find the true optimum, we can do better by searching in cuts of *decreasing* size over time.

We use step sizes that decrease by 1. From an initial step n , we drop at floor n , then $n + (n - 1)$, then $n + (n - 1) + (n - 2)$, and so on, running a linear scan from the last passed floor whenever the egg breaks. Each band is one floor shorter than the last, so a broken first egg always leaves the worst-case total at n . The floors this covers is the triangle number $\frac{n(n+1)}{2}$, so we want the smallest n with

$$\frac{n(n+1)}{2} \geq 100$$

Which is $n = 14$. Thus, the worst case is 14 drops.

Intuitively, the decreasing steps work by *spreading* the worst case evenly across every threshold placement rather than letting it collapse onto one. In the fixed-jump attempt the cost concentrated on a single floor (99), which sat as a spike above the rest. The decreasing schedule flattens that spike, so every placement costs exactly n , and the worst case drops to meet the average. This is the general shape of a minimax problem: whenever the maximum is uneven, the tall placement has slack elsewhere to borrow from, and we keep leveling until no placement can be improved without worsening another.

Problem 2.21

1024 pirates stand in a circle. They start shooting alternately in a cycle such that the 1st pirate shoots the 2nd, the 3rd shoots the 4th and so on. The pirates who got shot are eliminated from the game. They continue in circles, shooting the next standing pirate, till only one pirate is left. Which position should someone stand to survive?

Solution. Each pass around the circle, the number of pirates decreases by a factor of 2. Since the initial total is a power of 2, every pass starts with an even number of pirates. With an even count the pirates split exactly into shooter-victim pairs, so the last pirate of the pass is a victim, and the next shooter wraps back around to position 1. Thus position 1 shoots every pass and is never shot, until they are the last one remaining. So the pirate in position 1 survives.

Problem 2.22

I bought two fair six-sided dice. Keeping the first one untouched, I modified the second die, by erasing each face and writing a number of my own. Now, the sum of outcomes of the two dice is equally likely an integer between 1 and 12. Can you deduce the numbers written on the modified die?

Solution. The modified die has three zero faces and three six faces. The key detail is that any integer from 1 to 12 is equally likely, unlike the usual two-dice sum which peaks at 7 and tapers to a single way each of making 2 or 12. Since the standard die contributes at least 1, reaching a sum of 1 forces a 0 on the modified die, and reaching 12 forces a 6. The standard die alone is already uniform over 1 to 6, so we can build a uniform 1 to 12 by taking that block and shifting a copy of it up by 6. Adding 0 leaves the sum in 1 to 6, and adding 6 moves it to 7 to 12, so to weight the two blocks equally the modified die needs three 0 faces and three 6 faces.

Problem 2.23

Assume the gender of a child can be either Boy or Girl, equally likely.

- a. Suppose I have two children. Younger one is a girl. What is the probability that both children are girls?
- b. Suppose I have two children, and atleast one of them is a boy. Find the probability that both of them are boys.
- c. Suppose I have two kids, lets call them X and Y. Suppose X is a girl. What is the probability that I have two daughters?

Solution. The three parts differ in whether the condition names a specific child or only makes a claim about the pair. Naming a specific child leaves the other as an independent fair coin. An existential “at least one” condition does not.

- a. We condition on a specific child, the younger, being a girl. Writing each outcome as (older, younger), the condition leaves the two equally likely cases $\{(B, G), (G, G)\}$, of which (G, G) is both girls. So the probability is $\frac{1}{2}$, which is just the independent chance that the older child is also a girl.
- b. Saying *at least* one is a boy names no particular child, so it removes only (G, G) . This leaves three equally likely cases $\{(B, G), (G, B), (B, B)\}$, of which only (B, B) is both boys, giving $\frac{1}{3}$. The two mixed cases both satisfy the condition, which is why the denominator is 3 rather than 2.
- c. Labeling the children and telling us X is a girl fixes a specific child, exactly as in (a). The other child is an independent fair coin, so the probability of two daughters is $\frac{1}{2}$.

Problem 2.24

In a dark room, there is a deck of 52 cards, with exactly 10 cards facing up, rest are facing down. You can touch the cards with your hands but cannot sense which cards are facing up/down. You need to split all these cards into two decks with equal number of cards facing up.

Solution. We first note that the two decks only need an *equal* number of face-up cards between them, not a count matching the original 10. The two decks also need not be the same size.

Using this, we draw 10 cards at random, matching the number of face-up cards in the deck. Suppose this pile contains x face-up cards, so the remaining 42 cards hold the other $10 - x$ face up. Since our pile has 10 cards total, it also holds $10 - x$ face-down cards. We then flip every card in the pile, which turns those $10 - x$ face-down cards up and the x face-up cards down, leaving exactly $10 - x$ face up. Both piles now show $10 - x$ face-up cards.

Problem 2.25

A certain town comprises of 100 married couples. Some husbands secretly cheat on their wives. All wives know about the nature of every husband except their own. When a wife concludes that her husband cheated, she kicks her husband into the street at midnight. All husbands remain silent about their secret. One day, the mayor of the town announces to the whole town that there is at least 1 cheating husband in the town. After announcement, no one talks, waiting for someone to get kicked. Till 9th night from announcement, no husband was kicked, but on the 10th night, some husbands got kicked out simultaneously. How many are they?

Solution. 10 husbands were kicked out. Each wife saw either 9 cheating husbands (her husband cheats) or 10 (her husband is faithful), so each knows her count is either correct or lower by 1.

A wife who sees c cheaters therefore knows every other wife sees at least $c - 1$, so no one can act before night c , and the earlier nights tell her nothing. She watches only night c . If those c are all of them, they are kicked that night. When night c passes with no one kicked, the count must instead be $c + 1$, so her own husband is the extra cheater, and she kicks him on night $c + 1$. The 10 cheaters' wives each saw 9, ignored everything up to night 9, and when it stayed quiet they kicked their husbands together on night 10.

Problem 2.26

A father claims that it snowed last night. His first daughter says that the probability of snowfall on any given night is $\frac{1}{8}$. His second daughter says that the father lies 5 out of every 6 times. Given that the father claims it snowed, what is the probability that it actually did?

Solution. We can use Bayes' Theorem directly to solve for $P(S)$, that snowfall occurred given C , the father claimed it snowed. Using $P(C|S) = 1 - \frac{5}{6} = \frac{1}{6}$:

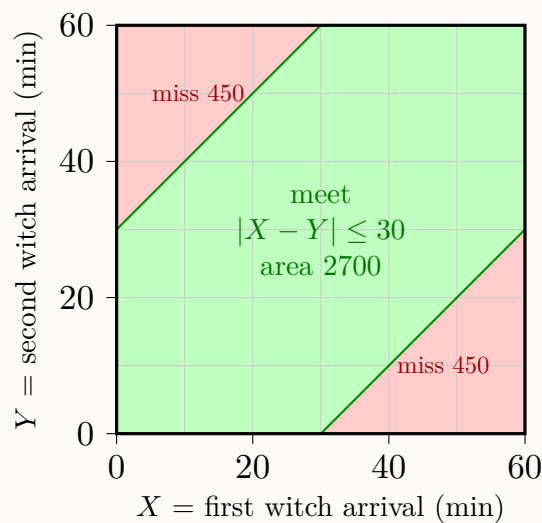
$$P(S|C) = \frac{P(C|S) \cdot P(S)}{P(C)} = \frac{\frac{1}{6} \cdot \frac{1}{8}}{\frac{1}{6} \cdot \frac{1}{8} + \frac{5}{6} \cdot \frac{7}{8}} = \frac{1}{36}$$

Problem 2.27

Two witches make a nightly visit to an all-night coffee shop. Each arrives at a random time between 0:00 and 1:00. Each one of them stays for exactly 30 minutes. On any one given night, what is the probability that the witches will meet at the coffee shop?

Solution.

The answer is $\frac{3}{4}$. If we plot each witches arrival as X and Y , then we can solve $|X - Y| \leq 30$:



Problem 2.28

Someone breaks into a king's wine cellar where he stores 1000 bottles of wine. This person proceeds to poison one of the 1000 bottles, but gets away too quickly for the king's guard to see which one he poisoned or to catch him.

The king needs the remaining 999 safe bottles for his party in 4 weeks. The king has 10 prisoners who deserve execution. The poison takes about 3 weeks to take effect, and any amount of it will kill whoever drinks it. How can he figure out which bottle was poisoned in time for the party?

Solution. We can use each prisoner's death as one bit of binary information. With 10 prisoners we can encode $2^{10} = 1024$ unique states, more than enough for 1000 bottles. We number the bottles in binary and assign each prisoner a bit position, and a prisoner drinks from a bottle whenever its bit is a 1. After 3 weeks, the set of prisoners who died reads off the poisoned bottle directly. For example, bottle 579 is 1001000011 in binary, so prisoners 1, 4, 9, and 10 drank from it, and they are the only ones who die.

Problem 2.29

Let X be a random variable that takes value 0 or 1 with 50% probability each. You need to define a new random variable Y as a function of X , such that Y has the value 1 with 25% probability and 0 otherwise.

Solution. A single draw of X can only give probabilities of 0, $\frac{1}{2}$, or 1, so we need two independent draws to reach $\frac{1}{4}$. We make Y the product of two independent instances of X :

$$Y = X_1 \cdot X_2$$

The product is 1 only when both draws are 1, which by independence happens with probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

3 Code

3.1 SQL Patterns (PostgreSQL)

Reference for recurring query shapes. Each entry states the problem phrasing that signals it and the mechanism that makes it work.

Reading a problem

Most prompts map to a small set of shapes, and the wording is the tell:

- “per group”, “for each” → **GROUP BY**, or a **PARTITION BY** window when the whole row must survive.
- “top- N per group”, “most recent per” → rank-then-filter, or **DISTINCT ON** when $N = 1$.
- “change from the previous”, “period-over-period”, “return” → **LAG** / **LEAD**.
- “running”, “cumulative”, “rolling”, “moving average” → an ordinary aggregate with **OVER**.
- “consecutive”, “streak”, “gap” → gaps and islands.
- “one column per category”, “pivot” → conditional aggregation.

The first decision on any prompt is whether the answer collapses rows (**GROUP BY**) or keeps every row and appends a computed column (a window function).

Logical processing order

A query is evaluated in this order, regardless of written order:

FROM/JOIN → **WHERE** → **GROUP BY** → **HAVING** → **SELECT** → **DISTINCT** → **ORDER BY** → **LIMIT**

Two consequences drive most of the patterns below. **WHERE** runs before aggregation, so it cannot see aggregates or **SELECT** aliases. Window functions are computed at the **SELECT** step, so they also cannot appear in **WHERE** and must be wrapped in a subquery or CTE to filter on.

3.1.1 Filtering and Joins

The foundation. Every later pattern assumes you can already restrict rows and combine tables.

WHERE vs HAVING. **WHERE** filters individual rows before grouping. **HAVING** filters whole groups after aggregation and is the only place an aggregate predicate is legal. A filter on a plain (non-aggregated) column belongs in **WHERE** so rows are dropped before the grouping work happens. Reserve **HAVING** for conditions like **COUNT(*) > 5**.

```
1 SELECT department, COUNT(*) AS headcount
2 FROM employees
3 WHERE active = true           -- row filter, before grouping
4 GROUP BY department
5 HAVING COUNT(*) > 5;         -- group filter, on an aggregate
```

Listing 1: **WHERE** prunes rows, **HAVING** prunes groups

Joins. The recognition cue is which side you want to keep when a match is missing.

- **INNER:** rows with a match on both sides.
- **LEFT / RIGHT:** keep all rows from one side, NULL-fill the other.
- **FULL OUTER:** keep unmatched rows from both sides.
- **CROSS:** Cartesian product, every left row against every right row.
- **SELF:** a table joined to itself under an alias (employee to manager, row to prior row).
- **Semi-join (EXISTS, IN):** keep left rows with at least one match. No row duplication, no right-side columns.
- **Anti-join (NOT EXISTS, or LEFT JOIN ... WHERE right.key IS NULL):** keep left rows with no match.

Semi- and anti-joins filter the left table on whether a match exists, without pulling right-side columns or duplicating rows. Prefer NOT EXISTS over NOT IN for the anti-join, since a NULL on the right poisons NOT IN to zero rows.

```
1  -- Semi-join: customers with at least one order
2  SELECT c.customer_id, c.name
3  FROM customers c
4  WHERE EXISTS (SELECT 1 FROM orders o
5                  WHERE o.customer_id = c.customer_id);
6
7  -- Anti-join: customers with no orders
8  SELECT c.customer_id, c.name
9  FROM customers c
10 WHERE NOT EXISTS (SELECT 1 FROM orders o
11                     WHERE o.customer_id = c.customer_id);
12
13 -- Anti-join, LEFT JOIN form: unmatched rows are the NULL-filled ones
14 SELECT c.customer_id, c.name
15 FROM customers c
16 LEFT JOIN orders o ON o.customer_id = c.customer_id
17 WHERE o.customer_id IS NULL;
```

Listing 2: Semi-join keeps matches, anti-join keeps non-matches

3.1.2 Aggregation

Conditional aggregation (pivot). The signal is a request for one column per category on a single row. Fold a categorical column into counts by wrapping a CASE in an aggregate. Rows the CASE does not match contribute 0 to SUM. In Postgres the FILTER clause does the same thing and reads more cleanly.

```
1  SELECT
2      inspection_type,
3      COUNT(*) FILTER (WHERE risk_category = 'Low Risk')      AS low_risk
4      ,
5      COUNT(*) FILTER (WHERE risk_category = 'High Risk')     AS
6      high_risk,
7      COUNT(*)                                                AS total
8  FROM sf_restaurant_health_violations
9  GROUP BY inspection_type
10 ORDER BY total DESC;
```

Listing 3: Pivot risk categories to columns

The `SUM(CASE WHEN ... THEN 1 ELSE 0 END)` form is equivalent and portable across engines. `COUNT` ignores `NULL`, so `COUNT(CASE WHEN cond THEN 1 END)` (no `ELSE`) also counts only the matches.

3.1.3 NULL Handling

`NULL` means unknown, so it propagates through arithmetic and poisons comparisons. The two functions below are the standard guards.

COALESCE and NULLIF. `COALESCE(a, b, c, ...)` returns the first non-`NULL` argument, evaluated left to right, or `NULL` if all are `NULL`. The common use is a fallback that keeps a `NULL` from poisoning arithmetic: `salary + commission` is `NULL` whenever `commission` is, so write `salary + COALESCE(commission, 0)`. It also supplies a default down a priority list, as in `COALESCE(nickname, first_name, 'Unknown')`. It is ANSI standard, unlike the vendor-specific `IFNULL` (MySQL) and `ISNULL` (SQL Server), and it accepts any number of arguments.

`NULLIF(a, b)` is the mirror image: it returns `NULL` when `a = b`, otherwise `a`. Its standard use guards division, `x / NULLIF(y, 0)` yields `NULL` instead of raising a divide-by-zero when `y` is 0.

```
1 SELECT
2     order_id,
3     revenue / NULLIF(quantity, 0) AS unit_price, -- NULL, not an
         error, when quantity = 0
4     COALESCE(discount, 0) AS discount -- NULL discount
         treated as 0
5 FROM orders;
```

Listing 4: `NULLIF` guards division, `COALESCE` supplies defaults

A subtlety with aggregates: `SUM`, `AVG`, and `COUNT(col)` skip `NULL`s. So `AVG(commission)` averages only the non-`NULL` rows, while `AVG(COALESCE(commission, 0))` counts the `NULL`s as zeros and enlarges the denominator.

Gotchas.

- **NOT IN with NULLs:** if the subquery returns any `NULL`, `x NOT IN (...)` is never true and the query returns zero rows. `x NOT IN (1, NULL)` expands to `NOT (x=1 OR x=NULL)`, and the `NULL` comparison poisons the result to unknown. Use `NOT EXISTS`, which is null-safe.
- **Integer division truncates:** `5 / 2` yields 2 because the fractional part is discarded. Cast one operand: `5::numeric / 2` or `5.0 / 2`. Bites hardest in ratio and percentage columns.
- **COUNT(col) vs COUNT(*):** the former skips `NULL`s in `col`, the latter counts every row. A common source of off-by-*N* errors.

3.1.4 Window Functions

The signal for a window function is a per-row result that still depends on other rows, such as a rank, a neighbor value, or a running total. Where `GROUP BY` would collapse the group to one row, a window function keeps every row and appends the computed column.

Every window function shares the same shape: `f() OVER (PARTITION BY ... ORDER BY ...)`. `PARTITION BY` splits the rows into independent groups, `ORDER BY` sets the order within

each group, and the function is evaluated per row. All of them are computed at the `SELECT` step, so to filter on their output you wrap them in a subquery or CTE.

Rank-then-filter with `ROW_NUMBER()`. Use this for top- N per group when you need to keep the full row, since `GROUP BY` would collapse the other columns away. Number rows inside each partition, then filter on that number in an outer query.

Choose the ranker by tie policy: `ROW_NUMBER` assigns distinct numbers and breaks ties arbitrarily, `RANK` repeats a rank and leaves gaps, `DENSE_RANK` repeats with no gaps.

```
1 SELECT department, name, salary
2 FROM (
3     SELECT department, name, salary,
4           ROW_NUMBER() OVER (PARTITION BY department
5                               ORDER BY salary DESC) AS rn
6     FROM employees
7 ) ranked
8 WHERE rn = 1;
```

Listing 5: Highest-paid employee per department

Offset access with `LAG()` and `LEAD()`. The signal is a comparison against the previous or next row: a delta, a period-over-period change, or a return. These read a value from another row relative to the current one, without a self-join. `LAG(col, k, default)` returns the value k rows earlier in the ordered partition, `LEAD` looks k rows ahead. Both default to $k = 1$ and return `NULL` past the partition edge unless a fill default is supplied. The `ORDER BY` inside `OVER` defines what “previous” means and is mandatory. `PARTITION BY` resets the sequence at each group boundary, so a lag never bleeds from one partition into the next.

```
1 SELECT ticker, trade_date, close,
2       close / LAG(close) OVER (PARTITION BY ticker
3                               ORDER BY trade_date) - 1 AS
4       daily_return
5 FROM prices;
```

Listing 6: Daily returns per ticker with `LAG`

The first row of each partition has no predecessor, so its `LAG` and the derived return are `NULL`, which is correct since there is no prior price to compare against. `LEAD` is the same tool aimed forward, used below for gap detection.

Running and rolling aggregates. The signal is “running”, “cumulative”, or “moving average”. An ordinary aggregate gains a window when you attach `OVER (... ORDER BY ...)`. With an `ORDER BY` and no explicit frame, the frame defaults to everything from the partition start up to the current row, giving a running total. Add an explicit `ROWS BETWEEN` frame to get a fixed moving window instead.

```
1 SELECT ticker, trade_date, close,
2       AVG(close) OVER (PARTITION BY ticker ORDER BY trade_date
3                       ROWS BETWEEN 19 PRECEDING AND CURRENT ROW) AS
4       ma_20,
5       SUM(volume) OVER (PARTITION BY ticker ORDER BY trade_date) AS
6       cum_volume
7 FROM prices;
```

Listing 7: 20-day moving average and running volume

`ma_20` is the current row plus the 19 before it, and `cum_volume` uses the default frame for a running total. These are the SQL forms of a pandas `.rolling(20).mean()` and `.expanding().sum()`.

One subtlety: the default frame is `RANGE UNBOUNDED PRECEDING`, which includes every row tied on the `ORDER BY` key, not just rows physically up to the current one. When ties on the order key exist and you want a row-precise window, state `ROWS BETWEEN` explicitly.

Gaps and islands. The signal is any mention of consecutive values, streaks, or runs. An *island* is a maximal run of consecutive values. The trick: subtract a dense row number from the value. Across a consecutive run both sides increase by one per step, so the difference stays constant, and it jumps exactly at a gap. Group by the difference to collapse each run.

```
1 WITH numbered AS (  
2     -- DISTINCT first so duplicate ids do not desync the row number  
3     SELECT DISTINCT log_id,  
4         ROW_NUMBER() OVER (ORDER BY log_id) AS rn  
5     FROM logs  
6 )  
7 SELECT MIN(log_id) AS island_start,  
8        MAX(log_id) AS island_end,  
9        COUNT(*)    AS length  
10 FROM numbered  
11 GROUP BY log_id - rn  
12 ORDER BY island_start;
```

Listing 8: Islands of consecutive ids

The `DISTINCT` is load-bearing. If `log_id` repeats, `rn` still advances on the duplicate while the value does not, so `log_id - rn` shifts and splits an island that should stay whole. For consecutive *dates* rather than integers, offset by an interval: `log_date - rn * INTERVAL '1 day'` is constant within a run.

```
1 SELECT log_id + 1 AS missing_from,  
2        next_id - 1 AS missing_to  
3 FROM (  
4     SELECT log_id,  
5         LEAD(log_id) OVER (ORDER BY log_id) AS next_id  
6     FROM logs  
7 ) t  
8 WHERE next_id - log_id > 1;    -- values missing between two present ids
```

Listing 9: Gaps via LEAD

3.1.5 Deduplication

DISTINCT. `DISTINCT` compares the entire select list as a tuple. `SELECT DISTINCT a, b` returns every distinct *pair*, so a value in `a` can still repeat as long as the paired `b` differs. It is a whole-row operator with no way to say “dedupe on this column and keep the rest of the row.” It runs after `SELECT` and before `ORDER BY`, which is why every `ORDER BY` expression must also appear in the select list, and it treats two `NULL`s as duplicates. `COUNT(DISTINCT col)` is a separate mechanism that dedupes the input to one aggregate and has no effect on the rows returned.

DISTINCT ON. A PostgreSQL extension that keeps the first row of each group sharing the same value of the parenthesized expressions, returning all selected columns of that surviving row. “First” is defined by `ORDER BY`, whose leading terms must match the `DISTINCT ON` expressions. Omit the `ORDER BY` and the surviving row is arbitrary.

```
1 SELECT DISTINCT ON (ticker)
2     ticker, trade_date, close
3 FROM prices
4 ORDER BY ticker, trade_date DESC;    -- leading term must be ticker
```

Listing 10: Most recent close per ticker

This is the compact form of the rank-then-filter pattern from the window section. The `ROW_NUMBER` equivalent needs a subquery and an outer `WHERE rn = 1`. Which to reach for:

- **Top-1 per group in Postgres:** `DISTINCT ON` is shorter and generally at least as fast, since it stops at the first row of each group.
- **Top-*N* for *N* > 1, or portability:** `ROW_NUMBER` extends to `rn <= 3` and runs on any engine. `DISTINCT ON` is Postgres-only.
- **Ties:** `DISTINCT ON` always emits exactly one row per group, settled by the remaining `ORDER BY` terms or arbitrarily. To keep every tied row, use `RANK() = 1`.

Because the `ORDER BY` is dictated by the deduplication key, presenting the result in a different order means wrapping the query in an outer `SELECT ... ORDER BY`.

3.1.6 String Manipulation

Positions are 1-based, and every function returns a new value rather than mutating in place. The recurring uses are normalizing a dirty join key, parsing a delimited field, and folding a group of strings into one.

Concatenation. The operator `a || b` propagates `NULL`: if either side is `NULL` the whole expression is `NULL`. `CONCAT(a, b, ...)` treats `NULL` as an empty string. `CONCAT_WS(sep, ...)` takes the separator first, joins the rest with it, and skips `NULL` arguments without leaving a doubled separator. Choosing `||` for user-facing text is a common way to silently blank out a whole column.

Case and whitespace. `UPPER` and `LOWER` are the standard folds, `INITCAP` capitalizes the first letter of each word and lowercases the rest. `TRIM` strips characters from the ends only, spaces by default, with `LEADING`, `TRAILING`, or `BOTH` to pick a side. Interior whitespace survives, so collapsing double spaces needs `REGEXP_REPLACE(s, '\s+', ' ', 'g')`. `LPAD(s, n, fill)` and `RPAD` pad to a fixed width and truncate when the input is already longer than `n`.

```
1 SELECT
2     LOWER(TRIM(email))                AS email_key,
3     SPLIT_PART(LOWER(TRIM(email)), '@', 2) AS domain,
4     LPAD(account_id::text, 8, '0')      AS padded_id,
5     CONCAT_WS(' ', last_name, first_name) AS sort_name
6 FROM users;
```

Listing 11: Normalizing a join key

Extraction and search. SUBSTRING(s FROM a FOR b) (or SUBSTR(s, a, b)) takes b characters starting at position a. LEFT(s, n) and RIGHT(s, n) take from the ends, and a negative n drops that many characters from the opposite end. STRPOS(s, sub) returns the 1-based index of the first occurrence and 0 when the substring is absent, so a found/not-found test compares against 0. SPLIT_PART(s, delim, n) pulls the nth field of a delimited string and returns an empty string when that field does not exist, worth wrapping in NULLIF(..., "").

```
1 SELECT full_name,
2        SPLIT_PART(full_name, ' ', 1) AS first_name,
3        NULLIF(SPLIT_PART(full_name, ' ', 2), '') AS last_name,
4        LENGTH(full_name)
5        - LENGTH(REPLACE(full_name, ' ', '')) AS space_count
6 FROM people;
```

Listing 12: Parsing a delimited name field

The last expression is the occurrence-counting idiom: remove every copy of the target and measure how much the string shrank. For a needle longer than one character, divide the difference by LENGTH(needle).

Replacement. REPLACE(s, from, to) substitutes every literal occurrence. TRANSLATE(s, from, to) maps character to character by position and deletes any character in from that has no counterpart in to, a fast way to strip a known character set. REGEXP_REPLACE(s, pattern, repl, flags) handles the general case and replaces only the *first* match unless the 'g' flag is passed, a frequent source of half-cleaned output.

Pattern matching. LIKE uses % for any run of characters and _ for exactly one, and it is case sensitive. ILIKE is the case-insensitive form. For real expressiveness the POSIX operators apply: ~ matches, ~* matches case-insensitively, and !~ negates. Anchors ^ and \$ matter, since an unanchored pattern matches anywhere in the string.

```
1 SELECT phone,
2        REGEXP_REPLACE(phone, '[^0-9]', '', 'g') AS digits,
3        phone ~ '^\+?[0-9]{10,15}$' AS looks_valid,
4        email ILIKE '%@gmail.com' AS is_gmail
5 FROM contacts;
```

Listing 13: Regex cleaning and validation

Splitting and aggregating. STRING_AGG(expr, delim ORDER BY ...) collapses a group into one delimited string and is the string analogue of SUM. The ORDER BY lives inside the call, so the output order is deterministic. The inverse direction expands one row into many: UNNEST(STRING_TO_ARRAY(s, delim)) or REGEXP_SPLIT_TO_TABLE(s, pattern) in the FROM clause.

```
1 SELECT department,
2        STRING_AGG(name, ', ' ORDER BY name) AS members
3 FROM employees
4 GROUP BY department;
5 -- one row per tag stored in a comma-separated column
6 SELECT p.id, TRIM(tag) AS tag
7 FROM products p,
8        UNNEST(STRING_TO_ARRAY(p.tags, ',')) AS tag;
```

Listing 14: Collapse a group, then expand it back

3.1.7 Command Reference

Signatures are PostgreSQL. Optional arguments are shown in brackets.

<i>Command</i>	<i>Arguments</i>	<i>What it does</i>
WHERE	boolean predicate	Filters individual rows before grouping. Cannot see aggregates, window functions, or select aliases.
GROUP BY	column or expression list	Collapses rows to one output row per distinct key combination.
HAVING	boolean predicate	Filters whole groups after aggregation. The only legal home for an aggregate predicate.
DISTINCT	applies to the select list	Removes duplicate rows compared across the entire select list. Treats two NULLs as equal.
DISTINCT ON	(<i>expr</i> , ...)	Postgres. Keeps one row per distinct value of the expressions, chosen by the leading ORDER BY terms, which must match.
ORDER BY	<i>expr</i> [ASC DESC] [NULLS FIRST LAST]	Sorts the output. Postgres places NULLs last under ASC and first under DESC .
LIMIT / OFFSET	<i>n</i>	Caps the number of rows returned, or skips the first <i>n</i> .
WITH	<i>name AS (query)</i>	Common table expression. Names a subquery for reuse. WITH RECURSIVE walks hierarchies.
CASE	WHEN <i>cond</i> THEN <i>val</i> [ELSE <i>val</i>] END	Row-level branching. Returns the first match, or NULL when nothing matches and ELSE is absent.
FILTER	(WHERE <i>cond</i>) after an aggregate	Restricts which rows the aggregate consumes. Postgres shorthand for SUM(CASE ...) .
EXISTS / NOT EXISTS	subquery	Semi-join and anti-join. Null-safe, unlike IN and NOT IN .

Query clauses and structure

<i>Command</i>	<i>Arguments</i>	<i>What it does</i>
COUNT(*)	none	Counts every row, including all-NULL ones.
COUNT(col)	one column	Counts rows where <i>col</i> is non-NULL.
COUNT(DISTINCT col)	one column	Counts distinct non-NULL values.
SUM / AVG / MIN / MAX	one expression	Skip NULLs. AVG divides by the non-NULL count.
STRING_AGG	(<i>expr</i> , <i>delim</i> [ORDER BY ...])	Concatenates a group into one delimited string.
ARRAY_AGG	(<i>expr</i> [ORDER BY ...])	Collects group values into an array.
COALESCE	any number of values	Returns the first non-NULL argument, left to right.
NULLIF	two values	Returns NULL when the arguments are equal, otherwise the first. Guards division by zero.
GREATEST / LEAST	list of values	Largest or smallest value across columns within one row. Ignores NULLs.
IS [NOT] DISTINCT FROM	two values	Null-safe comparison that treats two NULLs as equal.
IS [NOT] NULL	one value	The only valid null test. = NULL is always unknown.

Aggregates and NULL handling

<i>Command</i>	<i>Arguments</i>	<i>What it does</i>
OVER	(PARTITION BY ... ORDER BY ... [frame])	Turns a function into a window function evaluated per row without collapsing the result set.
ROW_NUMBER()	none	Sequential number within the partition. Ties broken arbitrarily.
RANK()	none	Competition rank. Ties share a rank and leave gaps afterward.
DENSE_RANK()	none	Ties share a rank with no gaps.
NTILE(n)	bucket count	Splits the partition into n roughly equal buckets and labels each row.
LAG(col[, k, default])	k defaults to 1	Value k rows earlier in the ordered partition. NULL past the partition edge unless a default is supplied.
LEAD(col[, k, default])	same as LAG	Value k rows ahead.
FIRST_VALUE / LAST_VALUE	col	First or last value in the current frame. LAST_VALUE needs an explicit frame to see the full partition.
NTH_VALUE(col, n)	column, position	The n th value in the frame.
SUM / AVG / COUNT ... OVER	aggregate plus window spec	Running or rolling aggregate computed per row.
ROWS BETWEEN	a PRECEDING to b FOLLOWING , CURRENT ROW , UNBOUNDED	Physical frame counted in rows. Use it for fixed moving windows.
RANGE BETWEEN	same bound keywords	Value-based frame that includes every row tied on the ORDER BY key. This is the default frame.

Window functions and frames

<i>Command</i>	<i>Arguments</i>	<i>What it does</i>
 	two strings	Concatenation. Yields NULL if either side is NULL.
CONCAT	any number of values	Concatenation treating NULL as an empty string.
CONCAT_WS	separator, then values	Joins arguments with a separator and skips NULLs.
LENGTH / CHAR_LENGTH	one string	Character count. OCTET_LENGTH gives bytes.
UPPER / LOWER /INITCAP	one string	Case folding. INITCAP capitalizes each word and lowercases the rest.
TRIM	[LEADING TRAILING BOTH] [chars] FROM s	Strips characters from the ends only, spaces by default. LTRIM , RTRIM , BTRIM are shorthands.
LPAD / RPAD	(s, n [, fill])	Pads to width n, truncating when s is already longer.
LEFT / RIGHT	(s, n)	First or last n characters. A negative n drops that many from the opposite end.
SUBSTRING	(s FROM a FOR b) or SUBSTR(s, a, b)	Slice of length b starting at 1-based position a.
STRPOS / POSITION	(s, sub) / (sub IN s)	1-based index of the first occurrence, 0 when absent.
REPLACE	(s, from, to)	Replaces every literal occurrence.
TRANSLATE	(s, from_set, to_set)	Maps characters by position and deletes those with no counterpart.
SPLIT_PART	(s, delim, n)	The nth delimited field. Returns an empty string when the field is missing.
STRING_TO_ ARRAY / UNNEST	(s, delim) / (array)	Splits into an array, then expands an array into rows.
REVERSE / REPEAT	(s) / (s, n)	Reverses a string, or repeats it n times.
LIKE / ILIKE	pattern with % and _	Wildcard match, case sensitive and case insensitive respectively.
~ ~* !~	POSIX regex	Regex match, case-insensitive match, and negated match.
TO_CHAR	(value, format)	Formats a number or timestamp as text, e.g. 'YYYY-MM-DD'.
CAST / ::	(expr AS type) / expr::type	Type conversion. ::numeric also fixes integer division.

String functions

3.2 Python Patterns

Good solutions to interview problems are often determined by the choice of container and knowledge of which built-in primitives express the required operation.

Containers and cost profiles

Selecting a container is selecting a cost profile. That decision precedes the algorithm and frequently dominates the runtime.

	Ordered	Mutable	Lookup	Use
<code>list</code>	yes	yes	$O(n)$	ordered sequence, duplicates permitted
<code>tuple</code>	yes	no	$O(n)$	fixed record, or a dictionary key
<code>set</code>	no	yes	$O(1)$	membership, deduplication
<code>dict</code>	insertion	yes	$O(1)$	mapping from key to value

Membership cost

`x in some_list` is $O(n)$. `x in some_set` is $O(1)$ on average. A membership test inside a loop over n elements is therefore $O(n^2)$ against a list and $O(n)$ against a set. Converting the list to a set once, before the loop, removes a factor of n . This substitution accounts for a large share of naive-to-optimal improvements.

Hashability. Set elements and dictionary keys must be hashable, which in practice requires immutability. A `tuple` is a valid key. A `list` is not, since mutation would invalidate its position in the hash table. Converting a list to a tuple to form a composite key is therefore a standard construction, as in grouping anagrams by their sorted letters.

```
1 seen = set()
2 seen.add((row, col))           # legal, tuple is hashable
3 # seen.add([row, col])         # TypeError: unhashable type
```

Listing 15: Tuple as a composite key

Dictionary access. Counting, grouping, and seen-before problems reduce to a dictionary. Three access forms differ in their treatment of a missing key.

```
1 d[k]                           # raises KeyError if k is absent
2 d.get(k, default)               # returns default instead of raising
3 d.setdefault(k, [])             # inserts default if absent, then returns it
4
5 for k in d:                     # iterates over keys
6 for k, v in d.items():           # iterates over pairs
```

Listing 16: Dictionary access

Built-in functions

An accumulator loop is often a built-in written out longhand. The following cover most cases.

Built-in	Expresses
<code>len, sum, min, max</code>	size and aggregation
<code>sorted</code>	ordering, parameterized by key and reverse
<code>any(...)</code>	existential quantifier: at least one element satisfies p
<code>all(...)</code>	universal quantifier: every element satisfies p
<code>enumerate</code>	index paired with value
<code>zip</code>	parallel iteration over several sequences
<code>reversed, range</code>	traversal order
<code>abs, round, divmod</code>	numeric utilities

`any` and `all` short-circuit. The expression `any(x < 0 for x in xs)` terminates at the first negative value.

The key argument. `sorted`, `min`, and `max` accept a key function applied to each element. Ranking is performed on the image of that function while the original element is returned. The consequence is that a record can be ranked on one of its fields and recovered whole.

```
1 min(scores)                                # returns a number
2 min(records, key=lambda r: r[1])           # returns the full record
3
4 sorted(words, key=len)                     # rank by length
5 sorted(records, key=lambda r: (-r[1], r[0])) # score desc, name asc
```

Listing 17: Ranking on a field, returning the record

A tuple key is the analogue of a multi-column `ORDER BY`. Negating a single numeric field reverses that field alone. `reverse=True` cannot express this, since it inverts every field simultaneously.

Comprehensions

A comprehension expresses selection and projection in one expression. The form `[f(x) for x in xs if p(x)]` corresponds to selecting $f(x)$ over the elements for which $p(x)$ holds. Four bracket types produce four containers.

```
1 [x*x for x in xs if x > 0]                # list
2 {x*x for x in xs}                        # set, deduplicates
3 {k: v for k, v in pairs}                  # dict
4 (x*x for x in xs)                        # generator, lazy, builds no intermediate list
```

Listing 18: The four forms

The generator form is appropriate when the result is consumed once, for example inside `sum(...)` or `max(...)`, since it avoids materializing a list. Nesting beyond two levels is generally less readable than an explicit loop.

Sorting

- `sorted(xs)` returns a new list. `xs.sort()` sorts in place and returns `None`. The assignment `xs = xs.sort()` binds `None` and fails silently.
- Python's sort is **stable**. Elements comparing equal retain their original relative order. Stability licenses sorting in multiple passes: sorting by the secondary field first and by the primary field second preserves the secondary ordering within each group.
- Sorting a sequence of lists or tuples compares elementwise from the left. The result depends on field order, so an explicit **key** should be supplied whenever the intent is nonobvious.
- Sorting costs $O(n \log n)$, which is negligible at the input sizes these problems use.

The collections module

Four types cover nearly every use. Three are specializations of the dictionary that eliminate the same recurring boilerplate: the check for whether a key already exists before writing to it. The fourth replaces the list when insertion or removal happens at the front.

Counter: frequency. A `Counter` is a dictionary from element to count. It is constructed in one pass over any iterable and answers the question “how many times does each value occur.” Frequency is the most common intermediate object in string and array problems, and constructing one by hand requires an existence check on every element.

```
1 from collections import Counter
2
3 c = Counter("mississippi") # Counter({'i': 4, 's': 4, 'p': 2, 'm': 1})
4 c['s']                    # 4
5 c['z']                    # 0, a missing key reads as zero and does NOT raise
6
7 c.most_common(2)          # [('i', 4), ('s', 4)], top 2 by count
8 c.most_common()          # every pair, ordered by count descending
9 c.total()                # sum of all counts
```

Listing 19: Counter

A missing key evaluates to 0 rather than raising `KeyError`, which is what permits `c[x] += 1` to be written unconditionally. The read does not insert the key, so the counter does not grow through inspection.

`most_common(k)` selects the top k with a heap in $O(n \log k)$. Calling `most_common()` with no argument sorts the whole counter at $O(n \log n)$, so the argument should be supplied whenever k is known.

Counters support multiset arithmetic, which expresses several problems directly.

```
1 a, b = Counter("aabbcc"), Counter("abbbd")
2
3 a + b                    # counts added
4 a - b                    # counts subtracted, results <= 0 are dropped
5 a & b                    # intersection: elementwise minimum
6 a | b                    # union: elementwise maximum
```

Listing 20: Counter arithmetic

Two canonical Counter identities

Anagram test. Two strings are anagrams if and only if their counters are equal, so `Counter(s) == Counter(t)` settles the question in $O(n)$. Sorting both strings answers the same question in $O(n \log n)$.

Sliding-window matching. To locate every permutation of a pattern p inside a text t , maintain a counter over a window of length $|p|$ and compare it against `Counter(p)` after each shift. Incrementing the entering character and decrementing the leaving one updates the window in $O(1)$.

defaultdict: grouping. A `defaultdict` is constructed with a zero-argument factory. Accessing a missing key calls that factory, inserts the result, and returns it. The effect is to remove the initialization branch from every accumulation loop.

```
1 from collections import defaultdict
2
3 groups = defaultdict(list)           # missing key -> []
4 for name, dept in employees:
5     groups[dept].append(name)        # no existence check required
6
7 counts = defaultdict(int)           # missing key -> 0
8 adj     = defaultdict(set)          # missing key -> set(), an
    adjacency list
9 grid    = defaultdict(lambda: defaultdict(int)) # a nested, sparse 2D
    table
```

Listing 21: defaultdict

The factory is any callable taking no arguments. `int()` returns 0, `list()` returns [], and `set()` returns an empty set, which is why the bare type name is passed rather than a call to it. A nested default requires a `lambda`, since `defaultdict(defaultdict(int))` would pass an instance where a factory is expected.

`defaultdict(list)` performs a GROUP BY that omits the aggregation step. Every member of each group survives, which is the property that `ROW_NUMBER() OVER (PARTITION BY ...)` supplies in SQL. Building a graph's adjacency list is the same operation applied to edges.

Reading a defaultdict mutates it

Any access to a missing key inserts it. The expression `if d[k]:` therefore creates k with an empty default and the dictionary grows through inspection alone. This corrupts a subsequent iteration over `d.items()` and inflates `len(d)`.

For a read that must not write, use `k in d` or `d.get(k)`. Neither triggers the factory.

The alternative to a `defaultdict` is `d.setdefault(k, []).append(x)` on an ordinary dictionary. It produces the same result and constructs a fresh empty list on every call, including the calls where the key is already present. The `defaultdict` factory runs only on a miss.

deque: a queue. A `deque` is a double-ended queue backed by a doubly linked list of fixed-size blocks. Appending and popping at either end is $O(1)$. The relevant comparison is against a `list`, which stores its elements contiguously, so removing the front element requires shifting every remaining element left by one position.

Operation	list	deque
Append or pop at the right	$O(1)$	$O(1)$
Append or pop at the left	$O(n)$	$O(1)$
Index at position i	$O(1)$	$O(n)$

The costs are complementary. A **list** is the correct stack, since a stack only touches the right end. A **deque** is the correct queue. Random access into a **deque** is a sign that the wrong container was chosen.

```

1 from collections import deque
2
3 q = deque([1, 2, 3])
4 q.append(4)           # right
5 q.appendleft(0)       # left
6 q.pop()              # from the right
7 q.popleft()          # from the left,  $O(1)$  where  $list.pop(0)$  is  $O(n)$ 
8 q.rotate(1)          # shift every element right by one, cyclically

```

Listing 22: deque

Breadth-first search is the standard application. The frontier is processed in insertion order, which requires removal from the front on every iteration of the main loop.

```

1 q = deque([start])
2 seen = {start}
3 while q:
4     node = q.popleft()      # FIFO, so nodes are visited by distance
5     for nxt in adj[node]:
6         if nxt not in seen:
7             seen.add(nxt)
8             q.append(nxt)

```

Listing 23: BFS frontier

Bounded length. A deque constructed with `maxlen=k` discards an element from the opposite end whenever a full deque is appended to. This maintains a rolling window over the last k items with no explicit eviction logic, which is useful for moving averages and for any statistic over a fixed lookback.

```

1 w = deque(maxlen=3)
2 for x in stream:
3     w.append(x)           # the oldest element is evicted automatically
4     if len(w) == 3:
5         print(sum(w) / 3)

```

Listing 24: A rolling window of the last 3 values

namedtuple: a labeled record. A tuple carries no information about what its positions mean, so `r[1]` requires the reader to remember the field order. A **namedtuple** attaches names while remaining a tuple, so it stays immutable, hashable, unpackable, and usable as a dictionary key or a set element.


```

1 from collections import namedtuple
2
3 Student = namedtuple("Student", ["name", "score"])
4 s = Student("Ada", 92.0)
5
6 s.score          # 92.0, and s[1] still works
7 name, score = s  # unpacking is unchanged
8 min(students, key=lambda s: s.score) # the key function reads clearly

```

Listing 25: namedtuple

The cost over a plain tuple is negligible and the gain is that a key function or a sort becomes self-documenting. When mutability is required, `dataclasses.dataclass` provides the equivalent with assignable fields.

Iteration

The following forms eliminate manual index arithmetic, which is the usual source of off-by-one errors.

```

1 for i, x in enumerate(xs):          # index and value together
2 for a, b in zip(xs, ys):            # parallel, halts at the shorter
    sequence
3 for a, b in zip(xs, xs[1:]):        # consecutive pairs
4 for x in reversed(xs):              # reverse traversal without index
    arithmetic
5
6 name, score = record                 # tuple unpacking
7 first, *rest = xs                    # starred unpacking

```

Listing 26: Iteration forms

The construction `zip(xs, xs[1:])` yields every adjacent pair and underlies run detection, difference sequences, and monotonicity checks. It is the analogue of `LAG` in SQL.

Problem structure

The four-move skeleton

A typical warm-up problem decomposes into four steps.

1. **Parse** the input into a list of records.
2. **Reduce** the records to a target key: a minimum, a maximum, a count, or the k -th distinct value.
3. **Filter** the records against that key.
4. **Order and format** the output.

Step 2 accounts for most errors. The characteristic failure is fusing steps 2 and 3 into a single pass that mishandles an edge case.

Ranking and ties. The most frequent error in step 2 concerns ties. Whether tied elements share a rank or consume one is the same decision in Python as in SQL.

Tie policy	Python	SQL
Ties consume a rank	<code>sorted(xs)[k-1]</code>	<code>ROW_NUMBER() = k</code>
Ties share a rank	<code>sorted(set(xs))[k-1]</code>	<code>DENSE_RANK() = k</code>

For scores [20, 20, 30], the second lowest is 30. Both elements equal to 20 hold rank 1 and neither occupies rank 2, so `sorted(xs)[1]` evaluates to 20 and is incorrect. The `set` converts ROW_NUMBER semantics into DENSE_RANK semantics.

Rank then filter. Once the key is identified, it is computed in one pass and the records are filtered against it in a second. The structure mirrors the necessity of wrapping a window function in a CTE, since a window function cannot appear in `WHERE`.

```
1 target = sorted({score for _, score in records})[1]    # rank
2 winners = sorted(name for name, score in records if score == target)
3                                     # filter, then order
```

Listing 27: Rank then filter

Common errors

- `sorted(set(xs))[1]` raises `IndexError` when all values are identical.
- Slicing copies. `xs[a:b]` costs $O(b - a)$, so a slice inside a loop makes the loop quadratic.
- `list.insert(0, x)` and `list.pop(0)` are $O(n)$, since every subsequent element shifts. A `deque` provides $O(1)$ operations at both ends.
- Integer division `//` floors toward negative infinity, so `-7 // 2` evaluates to `-4`. `int(-7 / 2)` truncates toward zero.
- A mutable default argument (`def f(xs=[])`) is created once at definition time and persists across calls. The default should be `None`, with the container constructed inside the func.
- Strings are immutable. Accumulating with `s += c` in a loop is $O(n^2)$. Collect the pieces in a list and apply `".join(parts)`.

3.3 Algorithmic Paradigms

Each paradigm is identified by a recognition cue in the problem statement.

- **Hashing.** Cue: find a pair, count occurrences, seen before, contains a duplicate. Space is traded for $O(1)$ lookup. The canonical construction tests whether the complement of the current element already appears in a map of previously visited elements.
- **Two pointers.** Cue: a sorted sequence, or a pairing problem. Pointers converging from both ends solve “two elements summing to t ” in $O(n)$ with constant extra space. Fast and slow pointers detect a cycle in a linked list.
- **Sliding window.** Cue: a contiguous subarray or substring subject to a constraint. The right edge expands to admit elements and the left edge contracts while the constraint is violated. Each element enters and leaves once, giving $O(n)$.
- **Prefix sums.** Cue: repeated range-sum queries over a static array. Cumulative sums are precomputed once, after which any range query is a difference of two entries. This is the array analogue of `SUM() OVER`.
- **Binary search on the answer.** Cue: the smallest x such that a condition is feasible. Requires a monotone predicate, meaning that feasibility at x implies feasibility at every larger x . The search ranges over the answer space rather than the input.
- **Sorting as preprocessing.** Cue: intervals, scheduling, or grouping equal elements. Sorting converts a global problem into a local one in which each element need only be compared against its neighbor.

Complexity anchors

Operation	Cost
<code>x in set, x in dict</code>	$O(1)$ average
<code>x in list</code>	$O(n)$
<code>list.append</code>	$O(1)$ amortized
<code>list.insert(0,x), list.pop(0)</code>	$O(n)$
<code>deque.appendleft, deque.popleft</code>	$O(1)$
<code>sorted(xs)</code>	$O(n \log n)$
<code>xs[a:b]</code>	$O(b - a)$, copies

3.4 Recursion and Memoization

A recursive function is defined by three obligations.

The recursion contract

1. **Base case.** At least one input for which the answer is returned without a recursive call.
2. **Recursive case.** The answer for input n expressed in terms of the answers for strictly smaller inputs.
3. **Progress.** Every recursive call moves the argument closer to a base case, so the recursion terminates.

Correctness follows by induction. Assuming the recursive call returns the correct answer on the smaller input, verify only that the combination step is correct. Tracing the full call tree by hand is unnecessary.

The call stack. Each pending call occupies a stack frame, so the space cost of a recursion is $O(d)$ where d is the maximum depth. Python imposes a default recursion limit of 1000 frames and raises `RecursionError` beyond it. A recursion whose depth scales with n is therefore unsafe for large n unless it is rewritten as a loop with an explicit stack.

Overlapping subproblems. Naive recursion recomputes identical subproblems. The Fibonacci recurrence $F(n) = F(n-1) + F(n-2)$ generates a call tree with $O(\varphi^n)$ nodes, since $F(n-2)$ is evaluated once by the $F(n-1)$ branch and again directly. Caching each result by its argument collapses the tree to $O(n)$ distinct evaluations.

```
1 from functools import lru_cache
2
3 @cache(maxsize=None)
4 def fib(n):
5     if n < 2:                                # base case
6         return n
7     return fib(n-1) + fib(n-2)               # recursive case
```

Listing 28: Memoization by decorator

Arguments to a memoized function must be hashable, which is the reason memoized state is passed as an integer, a string, or a tuple rather than as a list or a set.

Top-down and bottom-up. Memoized recursion is the top-down form. It computes only the states that are reachable, at the cost of a call stack. Tabulation is the bottom-up form. It fills an array in an order that guarantees every dependency is already resolved, using no stack and no recursion.

```
1 def fib(n):
2     if n < 2:
3         return n
4     prev, curr = 0, 1
5     for _ in range(n - 1):
6         prev, curr = curr, prev + curr    # only two states are ever live
7     return curr
```

Listing 29: The same recurrence, tabulated

The final line illustrates a general reduction. When a recurrence depends only on the previous k states, the table collapses from $O(n)$ space to $O(k)$.

Dynamic programming

Dynamic programming applies when a problem has **optimal substructure**, meaning the optimal solution is composed of optimal solutions to subproblems, together with **overlapping subproblems**, meaning those subproblems recur. Absent overlap, plain divide and conquer suffices. Absent optimal substructure, a greedy or exhaustive method is required.

Formulating a DP

Five questions, answered in order:

1. **State.** What arguments fully determine a subproblem? This is the signature of the function or the index set of the table.
2. **Recurrence.** How is a state's value expressed in terms of smaller states?
3. **Base cases.** Which states are known outright?
4. **Order.** In what order can states be evaluated so that dependencies are resolved first? Top-down recursion answers this automatically.
5. **Answer.** Which state holds the final result?

The complexity is the number of states multiplied by the cost of evaluating one state.

Canonical state definitions. Most problems reuse a small number of state shapes.

State	Meaning	Typical problem
$\text{dp}[i]$	best over the prefix ending at i	maximum subarray, house robber
$\text{dp}[i][j]$	best over two prefixes	edit distance, longest common subsequence
$\text{dp}[i][w]$	first i items, capacity w	0/1 knapsack
$\text{dp}[i][j]$	best over the interval $[i, j]$	matrix chain, palindrome partition

Backtracking

Backtracking is recursion over a search tree with an explicit undo step. It enumerates permutations, subsets, and combinations, and prunes branches that cannot yield a valid solution.

```

1 def solve(path, choices, results):
2     if is_complete(path):
3         results.append(path[:]) # copy, since path is mutated in place
4         return
5     for c in choices:
6         if not is_valid(path, c):
7             continue           # prune
8         path.append(c)         # choose
9         solve(path, remaining(choices, c), results)
10        path.pop()             # undo

```

Listing 30: Backtracking skeleton

The `path[:]` copy is mandatory. Appending `path` itself stores a reference to a list that subsequent mutation will alter, so every recorded result ends up identical.

For unconstrained enumeration, the standard library is preferable to a hand-written recursion.

```

1 from itertools import permutations, combinations, product
2
3 permutations(xs, r)      # ordered arrangements of length r
4 combinations(xs, r)     # unordered selections of length r
5 product(xs, repeat=r)   # r-tuples with repetition, the r-fold
                        Cartesian power

```

Listing 31: Enumeration without recursion

These correspond exactly to the counting formulas $P(n, r) = n!/(n - r)!$, $\binom{n}{r}$, and n^r .

3.5 Heaps and Priority Queues

A binary heap maintains the minimum of a collection under insertion and deletion at logarithmic cost. Python's `heapq` operates on an ordinary list interpreted as a heap, so no dedicated type is involved.

Operation	Call	Cost
Convert a list in place	<code>heapify(xs)</code>	$O(n)$
Insert	<code>heappush(h, x)</code>	$O(\log n)$
Remove the minimum	<code>heappop(h)</code>	$O(\log n)$
Inspect the minimum	<code>h[0]</code>	$O(1)$
Push then pop	<code>heappushpop(h, x)</code>	$O(\log n)$, one pass

A heap provides no ordering beyond its root. Iterating over the underlying list does not produce sorted output.

Minimum only. `heapq` implements a min-heap exclusively. A max-heap is obtained by negating the keys on insertion and negating again on extraction. For records, push the tuple `(-priority, item)`, since tuples compare lexicographically and the first field dominates.

Top- k under a stream

To retain the k largest of n elements, maintain a min-heap of size k . Each new element is compared against the root, which is the smallest of the current k . If the element is larger, the root is evicted and the element inserted.

The cost is $O(n \log k)$ and the memory is $O(k)$. Sorting the full input costs $O(n \log n)$ and requires all n elements in memory at once, so the heap is superior when $k \ll n$ and is the only option when the input arrives as a stream of unknown length.

```
1 import heapq
2
3 def top_k(stream, k):
4     h = []                                     # min-heap holding the k best
5     so far
6     for x in stream:
7         if len(h) < k:
8             heapq.heappush(h, x)
9         elif x > h[0]:                         # h[0] is the weakest survivor
10            heapq.heapreplace(h, x)           # pop the root, then push x
11    return sorted(h, reverse=True)
```

Listing 32: The k largest elements

When the whole input already fits in memory, the library provides this directly as `heapq.nlargest(k, xs)` and `heapq.nsmallest(k, xs)`, both of which accept a `key` argument.

Two heaps for a running median. Maintain a max-heap over the lower half of the values and a min-heap over the upper half, rebalancing after each insertion so their sizes differ by at most one. The median is then read from the root of the larger heap, or is the mean of the two roots when the sizes are equal. Insertion costs $O(\log n)$ and the median query is $O(1)$.

3.6 Sliding Window

A sliding window is a contiguous span of a sequence, delimited by two indices ℓ and r , together with a state summarizing the elements inside it. The window advances by admitting the element at r and evicting the element at ℓ . The technique applies when the state can be updated in $O(1)$ under those two operations, so that recomputing it from scratch is never necessary.

The recognition cue is a request for a *contiguous* subarray or substring subject to a constraint. Contiguity is required. A problem about subsequences or subsets admits reordering and belongs to a different paradigm.

Why the cost is linear

The amortization argument

Both indices are monotone. r advances once per iteration of the outer loop and ℓ never decreases, so each index takes at most n values over the entire run. The inner **while** loop is therefore bounded by n increments *in total* rather than n increments per outer iteration. The nesting is syntactic. The work is $O(n)$: every element is admitted exactly once and evicted at most once. The naive alternative enumerates all $O(n^2)$ spans and pays the cost of the state on each.

The precondition is that ℓ may safely move forward. This holds when the constraint is **monotone in the window**: if a window violates the constraint, every window containing it also violates it. Under that property, once $[\ell, r]$ fails, no window starting before ℓ and ending at r or later can succeed, so the discarded prefixes are provably useless.

Fixed-size windows

When the length k is given, there is no contraction step. The window is initialized once and then translated, one position at a time, by a paired admit and evict.

```
1 def max_window_sum(xs, k):
2     if len(xs) < k:
3         return None
4     total = sum(xs[:k])           # the window is built once, in O(k)
5     best = total
6     for r in range(k, len(xs)):
7         total += xs[r] - xs[r-k] # admit xs[r], evict xs[r-k]
8         best = max(best, total)
9     return best
10
11 max_window_sum([2, 1, 5, 1, 3, 2], 3) # 9, from the span [5, 1, 3]
```

Listing 33: Maximum sum over any window of length k

The single line `total += xs[r] - xs[r-k]` is the whole method. It replaces an inner loop of length k , reducing $O(nk)$ to $O(n)$. The index $r - k$ is the element leaving the window, since the window covering positions $r - k + 1$ through r has just replaced the one covering $r - k$ through $r - 1$.

A counter as the window state. The state need not be a number. Matching a pattern's permutations, described earlier under **Counter**, carries a frequency table as the state and compares it against a fixed target.


```

1 from collections import Counter
2
3 def find_anagrams(t, p):
4     k = len(p)
5     need = Counter(p)
6     have = Counter(t[:k])
7     out = [0] if have == need else []
8     for r in range(k, len(t)):
9         have[t[r]] += 1           # admit
10        have[t[r - k]] -= 1       # evict
11        if have[t[r - k]] == 0:
12            del have[t[r - k]]    # see the note below
13        if have == need:
14            out.append(r - k + 1) # the window's left index
15    return out
16
17 find_anagrams("cbaebabacd", "abc") # [0, 6]

```

Listing 34: Every starting index at which a permutation of p occurs in t

Each shift performs a constant amount of work and one dictionary comparison over an alphabet of bounded size, giving $O(n)$.

Zero-valued keys in a Counter

Decrementing to zero leaves the key present with value 0. Two consequences follow, and they behave differently.

Equality. On Python 3.10 and later, `Counter.__eq__` compares counts over the union of the keys and treats a missing key as zero, so `Counter({'a':1,'b':0}) == Counter({'a':1})` evaluates to `True`. On Python 3.9 and earlier the class inherits `dict.__eq__`, which compares key sets, and the same expression is `False`.

Size. `len(counter)` counts keys on every version. A key held at zero still contributes to `len`, so any window that tests the number of *distinct* elements must delete the key on reaching zero.

Deleting the exhausted key satisfies both, on every version.

Variable-size windows

When the length is unknown, r expands the window on every iteration and an inner loop moves ℓ forward. The two problem shapes differ in the test governing that inner loop and in the point at which the answer is recorded.

	Longest valid window	Shortest sufficient window
Inner loop	<code>while</code> invariant is broken	<code>while</code> condition holds
Effect	restore validity	test whether ℓ can be given up
Record at	after the loop, on a valid window	inside the loop, before evicting
Answer	<code>max(best, r - 1 + 1)</code>	<code>min(best, r - 1 + 1)</code>

Longest valid window. The window is repaired after each expansion, so it is valid at the bottom of every outer iteration, which is where its length is measured.

```

1 def longest_unique(s):

```

```

2     seen = set()                                # the state: which characters are in
        the window
3     left = best = 0
4     for right, c in enumerate(s):
5         while c in seen:                        # admitting c would break the
            invariant
6             seen.remove(s[left])               # evict from the left until it would
            not
7             left += 1
8             seen.add(c)                        # admit
9             best = max(best, right - left + 1)  # the window is valid here
10    return best
11
12    longest_unique("abcabcbb")                  # 3, from "abc"
13    longest_unique("pwwkew")                   # 3, from "wke"

```

Listing 35: Longest substring with no repeated character

The eviction loop is written before the admission so that the invariant is never violated inside the set. Admitting first and then repairing is equally correct provided the repair condition is written to match, and the ordering must be chosen once and held to.

```

1     from collections import defaultdict
2
3     def longest_k_distinct(s, k):
4         count = defaultdict(int)              # the state: a frequency table over
            the window
5         left = best = 0
6         for right, c in enumerate(s):
7             count[c] += 1                      # admit
8             while len(count) > k:              # invariant: at most k
                keys
9                 count[s[left]] -= 1
10                if count[s[left]] == 0:
11                    del count[s[left]]          # required, or len(count)
                is wrong
12                left += 1
13                best = max(best, right - left + 1)
14        return best
15
16    longest_k_distinct("eceba", 2)              # 3, from "ece"

```

Listing 36: Longest substring with at most k distinct characters

Here the admission precedes the repair, because the invariant is stated over the window's state rather than over the incoming element. The `del` is what makes `len(count)` equal to the number of distinct characters currently present.

Shortest sufficient window. The inner loop contracts while the target condition is still satisfied, which searches for the smallest left index that keeps it satisfied. The length is recorded before each eviction, since the window may cease to qualify immediately afterward.

```

1     def min_length(xs, t):
2         total = 0
3         left = 0
4         best = float('inf')

```

```

5     for right, x in enumerate(xs):
6         total += x                                # admit
7         while total >= t:                          # the window already
            suffices
8             best = min(best, right - left + 1)    # record, then try to
            shrink
9             total -= xs[left]                    # evict
10            left += 1
11    return 0 if best == float('inf') else best
12
13 min_length([2, 3, 1, 2, 4, 3], 7)    # 2, from [4, 3]
14 min_length([1, 1, 1], 7)            # 0, no window qualifies

```

Listing 37: Shortest subarray with sum at least t , over positive values

`float('inf')` as the initial value of a minimum removes the special case for an empty candidate set. The sentinel is then translated to whatever the problem specifies for “no such window,” which is commonly 0.

When the window does not apply

The monotonicity precondition

`min_length` above is correct only because every element is positive. Positivity makes `total` nondecreasing in the window’s length, so `total >= t` is monotone and a failing prefix can be discarded permanently.

Introduce a negative value and that reasoning collapses. Extending a window may now decrease its sum, and a left index abandoned earlier could belong to the optimal answer. On `[3, -2, 5]` with $t = 6$, the window `[3, -2, 5]` qualifies while neither the prefix `[3, -2]` nor the suffix `[5]` taken alone reveals it by contraction.

The replacement is prefix sums with a hash map, which considers every left endpoint rather than only the ones a monotone rule preserves. The check for a negative value in the constraints is therefore the first thing to perform on any subarray-sum problem.

Common errors

- The window length is `right - left + 1`. Omitting the `+1` is the standard off-by-one, since both endpoints are inclusive.
- Recording the answer at the wrong point. A longest-window problem measures after the repair loop. A shortest-window problem measures inside the contraction loop, before the eviction.
- An `if` where a `while` is required. One eviction restores the invariant only when a single admission can break it by exactly one unit. The general repair is a loop.
- Failing to evict from the state. Advancing `left` without removing `xs[left]` from the set, counter, or running sum leaves the state describing a window that no longer exists.
- Retaining zero-valued keys in a `Counter` or `defaultdict` while testing `len` against a distinct-count bound.
- Recomputing the state over the span, as in `sum(xs[left:right+1])` inside the loop. This restores the $O(n^2)$ cost the window exists to remove.